Flat in the Wrong Coordinates: Optimizer-Induced Symmetry Drift Biases Sharpness Comparisons Between Adam and SGD

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Abstract

Empirical studies often compare how flat the minima found by different optimizers are, and the conclusions disagree. A preliminary version of this study used a common relative random-perturbation metric and found Adam's minima *flatter* than SGD's on MNIST and CIFAR-10. We show that such comparisons mix two factors: the geometry of the learned function, and the point of the network's rescaling-symmetry orbit at which that function happens to be represented. For the Hessian trace, which in ReLU networks without normalization layers equals the Gauss–Newton trace, the split is exact: raw trace = *minimum sharpness* (the trace minimized over function-preserving neuron rescalings) × *orbit excess*. We prove that SGD's first-order drift along these orbits vanishes identically, whereas Adam's diagonal preconditioning moves it along them at first order. Empirically, at small learning rates, the drift scales approximately as $\eta\sqrt{\text{fan-in}}$. Adam, and likewise RMSprop and sign descent, keeps its orbit excess low (≈ 1.2 on MLPs, 1.4–1.8 on a CNN) across hyperparameters and training stages. SGD, which cannot move along the orbits at first order, and SGD with momentum and SAM, which move along them only weakly, sit at an excess of up to 3.4. In our experiments (162 loss-matched training runs of MLPs and a small CNN, up to 17 measures), this coordinate factor is as large as the functional differences being measured. On a CIFAR-10 MLP it favours Adam by about 2× and reverses the ordering: the probability that Adam is rated flatter than SGD is 0.62–0.75 under coordinate-dependent trace measures but 0.01–0.22 under minimum sharpness. On MNIST it flips the raw ordering in the middle of training. On a CNN, where Adam's solutions are clearly sharper in orbit-minimized sharpness, it hides most of the difference relative to SGD with momentum. Across MLPs with 1 to 8 hidden layers the coordinate factor stays at 1.4–1.8× while Adam's orbit-minimized sharpness grows with depth, so the same factor reverses the ordering in shallow networks and masks it in deep ones. A causal intervention undoes Adam's drift after every step with a rescaling that preserves the network function to float32 precision. It roughly doubles raw trace-type sharpness at matched loss (median ×1.9 on CIFAR-10), leaves minimum sharpness unchanged (×0.98), and removes Adam's apparent flatness advantage over SGD. In noisy regimes it also stalls training. Holding Adam at the minimum-trace point instead does not, while holding it at the minimum-norm point stalls it like Adam-Q, which suggests that in these settings the drift also helps Adam train. Moving SGD to the trace-minimizing coordinates instead (symmetry teleportation) destabilizes it: 10 of 16 teleported runs diverged, against none without teleports, because each teleport raised $\lambda_{\max}$ toward or above $2/\eta$. On long horizons with label noise, the orbit serves as a testbed for implicit sharpness regularization in a nonlinear network: SGD drifts toward the trace-minimizing point and Adam toward the minimizer of $\sum_i \sqrt{G_{ii}}$, the destinations predicted by existing implicit-regularization theory, in all 32 label-noise runs of each optimizer (16 independent initializations), and in a further 12 runs per optimizer the direction holds when the curvature is held fixed and only the motion along the orbit is evaluated. Started from different points of the same orbit, both optimizers lose the initial

difference in orbit excess within one epoch, yet Adam ends below SGD from every starting point. With BatchNorm (an MLP, and ResNet-18 on the full CIFAR-10 training set), raw measures follow weight norms set by the learning rate and weight decay, so their verdict depends on which hyperparameters are paired, while measures invariant to every per-weight rescaling give one verdict per network; that verdict rates Adam flatter in the MLP and sharper in ResNet-18. In these sweeps, removing the coordinate factor also raises the cross-optimizer correlation between trace-type sharpness and generalization (Kendall τ 0.38 $\rightarrow$ 0.61 on CIFAR-10, 0.47 $\rightarrow$ 0.62 on MNIST) while leaving within-optimizer correlations unchanged. Re-running our original protocol shows that its "Adam is flatter" result came from this coordinate factor on CIFAR-10 and from a training-loss mismatch on MNIST. We recommend that cross-optimizer flatness comparisons report rescaling-invariant measures, or at least the orbit excess.

1 Introduction

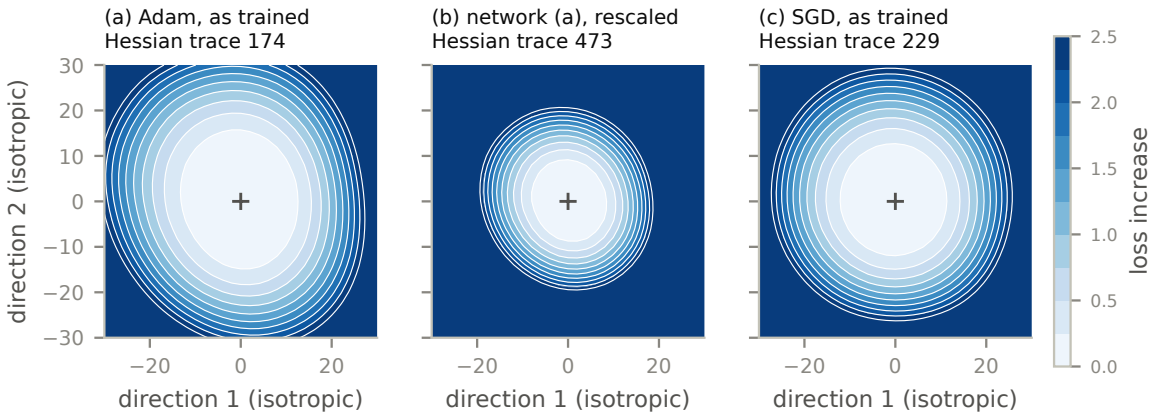
The idea that flat minima generalize better than sharp ones [15, 21] motivates both analysis tools and optimizers such as SAM [14]. A recurring empirical question is *which optimizer finds flatter minima*. Adaptive methods such as Adam [23] are often reported to generalize worse than SGD [43], and a popular explanation is that they converge to sharper minima [47]. Measurements nevertheless disagree, and recent theory suggests that Adam and SGD implicitly minimize *different* notions of sharpness [29].

This paper began as an attempt to answer that question with a standard tool, the relative increase in loss under isotropic Gaussian weight noise, $(L(w + \sigma\varepsilon) - L(w))/L(w)$. The preliminary result was that Adam's minima appeared *flatter* than SGD's on both MNIST and CIFAR-10, the opposite of the usual narrative. Before reporting it, we checked whether it was an artifact. It was, and the reason turned out to be general and mechanistic.

Coordinate-dependent sharpness is known to be manipulable: Dinh et al. [12] showed that function-preserving reparameterizations of ReLU networks can make any minimum arbitrarily sharp or flat. A common reply is that no optimizer performs such reparameterizations, so the objection is academic. Our central finding is that **Adam does perform them**. It moves systematically along the network's rescaling symmetries and settles at coordinates whose Hessian trace is typically only 10–50% above the minimum attainable on the same rescaling orbit. SGD provably does not move along them at first order. A raw sharpness comparison between Adam and SGD therefore compares two functions *and* two positions on their symmetry orbits, and we show that the second factor is often the larger one. Figure 1 illustrates this on two trained CIFAR-10 networks.

Contributions.

- **A diagnostic decomposition** (§3). Writing the trace as the minimum sharpness of the function [18] times an orbit excess ≥ 1 is immediate from the definitions. We use it to attribute cross-optimizer differences in the trace to the function or to the coordinates, and show that for ReLU networks without normalization layers it applies to the Hessian trace itself, which there equals the Gauss–Newton trace (Lemma 2).
- **A mechanism with a proof and a scaling law** (§5.1). The first-order drift of the per-unit conserved quantity $\|w_{\text{in}}\|^2 - \|w_{\text{out}}\|^2$ vanishes identically for plain SGD, is second order for heavy-ball momentum, is proportional to the perturbation radius for SAM, and is generically nonzero for diagonally preconditioned methods (the non-equivariance itself is known, §2). For sign-like updates it scales with the units' ℓ_1 norms, which predicts the $\eta\sqrt{\text{fan-in}}$ law we measure (fitted exponents 0.91 and 0.94).



(a) and (b) compute exactly the same function; their landscapes differ only through the coordinates.

Figure 1: **Same function, different landscape.** Two-dimensional loss slices along the same two random isotropic directions around (a) an MLP trained with Adam, (b) the *identical function* obtained from (a) by rescaling each hidden neuron's incoming and outgoing weights (to the minimum-norm representative), and (c) an MLP trained with SGD. All three are at training loss ≤ 0.01 on CIFAR-10. By the isotropic picture, or its Hessian trace, Adam's solution looks flatter than SGD's, yet a function-preserving change of coordinates makes it look 2.7× sharper. Adam's training dynamics drift toward coordinates like (a); SGD's cannot at first order (§5.1).

- **A loss-matched, multi-measure audit** (§5.2). We compare SGD, SGD+momentum, SAM and Adam at *matched training loss* on MLPs (MNIST, CIFAR-10) and a CNN (CIFAR-10), using 17 measures. Adam keeps a low orbit excess (≈ 1.2 on the MLPs, 1.4–1.8 on the CNN) while the other optimizers reach 3.4. Depending on the size of the true functional difference, this coordinate factor reverses the optimizer ranking (CIFAR-10 MLP), flips it during training (MNIST), or masks a large difference (CNN). It does not grow with depth (§5.9).
- **A causal test** (§5.3). *Adam-Q* applies, after each Adam step, the function-preserving rescaling that undoes the drift. It changes the coordinates but not the function at the moment it is applied. At matched training loss raw sharpness roughly doubles while invariant sharpness is unchanged. In noisy regimes Adam-Q also stalls. Holding Adam at the minimum-trace point does not, and holding it at the minimum-norm point does, so in this MLP the drift also matters for optimization.
- **A two-phase picture of orbit dynamics** (§5.1, §5.4, §5.5). Adam's first-order drift takes it close to the trace-minimizing point within a few epochs. A slower, noise-driven drift then moves SGD toward minimum trace and Adam toward minimum $\sum_i \sqrt{G_{ii}}$, which we confirm with label noise in all 32 runs of each optimizer, from 16 independent initializations, and with a test that holds the curvature fixed. Starting the same function at different orbit points shows that the excess is set by training dynamics rather than inherited from initialization, and that Adam's lower excess holds from every start. Adam's coordinates do not help SGD: teleporting SGD to them raises $\lambda_{\max}$ toward $2/\eta$, and most teleported runs diverge.
- **Consequences** (§5.6, §5.7, §5.10). The coordinate factor specifically degrades *cross-optimizer* sharpness–generalization correlations, and it explains (together with a loss-level confound) our original result. With BatchNorm (an MLP and ResNet-18 on full CIFAR-10), raw measures are

dominated by weight norms set by the learning rate and weight decay, and the raw ordering is set by the hyperparameter pairing while invariant measures give one verdict per network.

2 Related work

Sharpness and generalization. Many measures exist: worst-case and average-case perturbation sharpness [19, 21], the Hessian trace and top eigenvalue, and adaptive or normalized variants designed for scale invariance [26, 28, 36, 37, 41]. Large-scale studies find that their correlation with generalization is fragile [2, 19, 20]. Dinh et al. [12] exhibited reparameterizations that change sharpness arbitrarily without changing the function. Kristiadi et al. [24] show more generally that Hessian-based flatness measures are not invariant under reparametrization unless the metric is represented explicitly, and Ahmed et al. [1] define sharpness in the Fisher geometry so that it is invariant to smooth function-preserving reparametrizations. Ibayashi et al. [18] proposed *minimum sharpness*, the minimal Hessian trace over rescaling-equivalent parameterizations, and reported that it correlates well with generalization. We use their quantity, generalize it to other powers of the curvature and to conv nets, and show that the gap between raw trace and minimum sharpness is systematically *optimizer-dependent*. Zhang et al. [44] observed that Adam can break the correlation between flatness and generalization, and proposed a parameterization-invariant Bayesian alternative. We identify a concrete mechanism behind such breakdowns and measure its size.

Symmetries and conservation laws. Gradient flow on homogeneous networks conserves $\|w_{\text{in}}\|^2 - \|w_{\text{out}}\|^2$ per unit [13]. Path-SGD [35] was designed to make optimization itself invariant to these rescalings. Kunin et al. [25] analysed how finite learning rates, momentum and weight decay break such laws. Marcotte et al. [30, 31] characterized conservation laws for gradient flows and showed that none survive momentum dynamics in ReLU networks. Ziyin et al. [48, 49] showed that SGD's minibatch noise slowly moves parameters along rescaling and other exponential symmetries toward a "noise equilibrium". This is a second-order effect, consistent with Proposition 1a. Closest to our mechanism, Singh [40] showed that gradient descent, momentum, Shampoo and Muon are equivariant under the gauge symmetry of matrix factorization while coordinate-wise methods (Adam, RMSProp, signum) are not, and that this changes which interpolant is found. Our Proposition 1 is the analogous statement for the ReLU rescaling symmetry. Symmetry *teleportation* [45, 46] moves parameters along orbits deliberately, to speed up optimization or to reach minima of different curvature, and Lebeurrier et al. [27] choose a rescaling of a ReLU network that speeds up training. Saul [38] rebalances ReLU networks by per-unit multiplicative rescalings that minimize $\ell_{p,q}$ weight norms; our coordinate descent for MS_p is the analogous procedure for curvature instead of norms. Medeiros [33] analyses how discrete gradient descent breaks the ReLU conservation laws at second order, near the edge of stability, which is the $D^{(2)}$ term of Proposition 1a. Three recent preprints are closest to our mechanism. Shirodkar [39] reports that Adam's per-coordinate preconditioner drifts along symmetry orbits, including the ReLU rescaling, and constructs gauge-equivariant variants of Adam. Kim [22] shows that no standard optimizer is equivariant to Transformer reparameterizations, and that a function-preserving rescaling that unbalances paired weight norms degrades later training even when AdamW's moment estimates are transformed, which parallels our Adam-Q and teleportation results (§5.3, §5.4) in a different architecture. Together with Singh [40], these establish that adaptive optimizers move along, and are affected by, the symmetry orbits. Our contribution is not that adaptive methods break these symmetries, which follows from the work above. It is to show where the drift leads (near minimum sharpness), to measure how large it is ($\eta\sqrt{\text{fan-in}}$; a factor ≈ 2 in the Hessian trace), and to show that it biases flatness comparisons between optimizers enough to

Table 1: Comparison with Li et al. [29], the closest work on optimizer-dependent sharpness. Entries for their paper summarize its main results; see their paper for the precise assumptions.

	Li et al. [29]	This paper
Question	Which sharpness does Adam implicitly reduce near a manifold of minimizers?	How much of a *measured* Adam–SGD sharpness difference comes from the position on a rescaling orbit?
Kind of result	Theorem, in a slow-SDE approximation (small learning rate, long horizon) near the minimizer manifold, with label noise	Exact per-step identity for the first-order orbit drift (Prop. 1); everything else is measured
Sharpness	$\operatorname{tr} \operatorname{Diag}(H)^{1/2}$ for Adam vs. $\operatorname{tr} H$ for SGD	17 measures; the trace split into $\mathrm{MS}_1 \times E$
Optimizers	Adam and other adaptive methods (RMSProp, Adam-mini, Adalayer) vs. SGD	SGD, SGD+M, SAM, Adam, RMSprop, signSGD
Experiments	Sparse linear regression with diagonal linear networks; deep matrix factorization (a failure case)	MLPs, a small CNN and a BatchNorm MLP on MNIST / CIFAR-10 subsets
Relation	—	We do not re-derive their result. Our label-noise runs (§5.5) use the orbit, where the minimizer of $\sum_i \sqrt{G_{ii}}$ is computable exactly, as an *empirical test* of their prediction in nonlinear networks

reverse, flip or mask them.

Implicit sharpness regularization. SGD with label noise implicitly minimizes $\operatorname{tr} H$ near the minimizer manifold [7, 10], and Li et al. [29] prove that Adam instead minimizes $\operatorname{tr} \operatorname{diag}(H)^{1/2}$. Gradient methods train at the edge of stability [8, 9], so their sharpness is largely set by the learning rate. We see this too (Appendix E), which is why we compare optimizers over many learning rates rather than at one.

Table 1 separates our contribution from Li et al. [29]. Their theorem concerns the slow, noise-driven dynamics near the minimizer manifold and says which sharpness Adam reduces. Our claims concern the fast, first-order drift along an exact symmetry, and what it does to sharpness *measurements*; the two are complementary, and our slow-phase results test theirs rather than extend it.

3 Rescaling orbits, a conservation law, and minimum sharpness

Consider a network built from linear (or convolutional) layers and positively homogeneous activations (ReLU, max-pooling). For a hidden unit j (a neuron, or a convolutional channel) let a_j collect its incoming weights and bias and c_j its outgoing weights. For $s \in \mathbb{R}$, the map $T_s^{(j)} : (a_j, c_j) \mapsto (e^s a_j,\, e^{-s} c_j)$ leaves the network function unchanged. Applying such maps to all units generates the *rescaling orbit* of θ.

Lemma 1 (Euler identity)**.** *For any loss ℓ that depends on θ only through the network function, including any minibatch loss, $\langle a_j, \nabla_{a_j} \ell \rangle = \langle c_j, \nabla_{c_j} \ell \rangle$ for every hidden unit j.*

Proof. $s \mapsto \ell(T_s^{(j)} \theta)$ is constant; differentiate at $s = 0$. □

Let $Q_j = \|a_j\|^2 - \|c_j\|^2$. Lemma 1 gives $\dot{Q}_j = 0$ under gradient flow [13]. For a discrete update $\theta^+ = \theta + u$, expanding $\|a + u_a\|^2$ gives the *exact* decomposition

$$Q_j(\theta^+) - Q_j(\theta) = \underbrace{2\big(\langle a_j, u_{a_j}\rangle - \langle c_j, u_{c_j}\rangle\big)}_{D_j^{(1)}\ \text{(first order)}} + \underbrace{\|u_{a_j}\|^2 - \|u_{c_j}\|^2}_{D_j^{(2)}\ \text{(second order)}}. \tag{1}$$

Proposition 1. ***(a) SGD.*** *For $u = -\eta\nabla\ell_B$ on any minibatch B, $D_j^{(1)} = 0$ exactly and $D_j^{(2)} = \eta^2(\|\nabla_{a_j}\ell_B\|^2 - \|\nabla_{c_j}\ell_B\|^2)$.* ***(b) Heavy-ball momentum.*** *For $u_t = -\eta\sum_{k\geq 0}\beta^k g_{t-k}$, $D_j^{(1)} = -2\eta\sum_k \beta^k\big(\langle a_{j,t} - a_{j,t-k}, g_{a,t-k}\rangle - \langle c_{j,t} - c_{j,t-k}, g_{c,t-k}\rangle\big)$. Each term contains a parameter difference, which is itself $O(\eta)$, so the drift is again second order in η.* ***(c) Diagonal preconditioning (Adam, RMSProp, sign descent).*** *For $u = -\eta P^{-1}m$ with diagonal $P \succ 0$, $D_j^{(1)} = -2\eta(\langle a_j, P_a^{-1}m_a\rangle - \langle c_j, P_c^{-1}m_c\rangle)$, which is generically nonzero. For sign descent, $u = -\eta\,\mathrm{sign}(g)$, this gives $D_j^{(1)} = 2\eta(\rho_a\|a_j\|_1 - \rho_c\|c_j\|_1)$, where $\rho_x = -\langle x, \mathrm{sign}(g_x)\rangle/\|x\|_1 \in [-1,1]$ is the ℓ_1-weighted net fraction of coordinates moving away from zero.* ***(d) SAM.*** *For $u = -\eta\,\nabla\ell_B(\theta+\varepsilon)$ with $\varepsilon = r\,\nabla\ell_B(\theta)/\|\nabla\ell_B(\theta)\|$, $D_j^{(1)} = 2\eta(\langle\varepsilon_{a_j}, \tilde{g}_a\rangle - \langle\varepsilon_{c_j}, \tilde{g}_c\rangle)$ with $\tilde{g} = \nabla\ell_B(\theta+\varepsilon)$: first order in η but proportional to the radius r. (With momentum, the terms of (b) add.)*

Proof. (a) Lemma 1 applied to ℓ_B. (b) Apply Lemma 1 at time $t-k$ to g_{t-k} and subtract. (c) Direct substitution. (d) Lemma 1 at $\theta+\varepsilon$ gives $\langle a_j + \varepsilon_{a_j}, \tilde{g}_a\rangle = \langle c_j + \varepsilon_{c_j}, \tilde{g}_c\rangle$; substitute. □

Scope. Proposition 1 is an exact statement about a single discrete step, not a continuous-time limit. It assumes that the loss depends on θ only through the network function, that a_j includes the unit's bias, and that there is no weight decay: an L_2 term $\frac{\lambda}{2}\|\theta\|^2$ adds $-2\eta\lambda Q_j$ to $D_j^{(1)}$ for SGD, so Q_j decays toward zero instead of being conserved. None of the MLP and CNN experiments uses weight decay. In (b), the parameter differences accumulate over the momentum horizon, so the prefactor of η^2 depends on β and on how gradients are correlated across that horizon; we do not bound it. In our runs SGD+M's total drift matches SGD's at equal effective learning rate $\eta/(1-\beta)$ (§5.1). The second-order term $D_j^{(2)} = \eta^2(\|g_{a_j}\|^2 - \|g_{c_j}\|^2)$ is not zero in general (it vanishes only when the incoming and outgoing gradient norms agree), so "SGD does not drift" always means "not at first order": SGD can drift slowly, and §5.5 studies where.

Remark 1 (Why the drift scales with $\sqrt{\text{fan-in}}$). SGD moves each weight in proportion to the magnitude of its partner weights, and this is what keeps Q_j fixed. Sign-like updates move every coordinate by about η whatever its magnitude. By Proposition 1c, $D_j^{(1)} = 2\eta(\rho_a\|a_j\|_1 - \rho_c\|c_j\|_1)$. If the two alignment fractions are comparable and positive ($\rho_a \approx \rho_c > 0$, norm growth on both sides), the side with the larger ℓ_1 norm gains squared norm faster; without that assumption the sign of the drift can reverse. For dense vectors $\|x\|_1 \approx \kappa\sqrt{d}\,\|x\|_2$ ($\kappa = \sqrt{2/\pi}$ for Gaussian entries), so a unit whose fan-in exceeds its fan-out drifts toward large incoming norm at a rate $\propto \eta\sqrt{d_{\text{in}}}$. Adam behaves like sign descent when gradient noise is small relative to the signal [4, 6].

Exact transport of curvature along the orbit. A rescaling acts linearly, $\theta' = D\theta$ with D diagonal and positive. The Hessian and the Gauss–Newton matrix G [32] therefore transform *exactly* as $H(\theta') = D^{-1}H(\theta)D^{-1}$, and likewise $G(\theta') = D^{-1}G(\theta)D^{-1}$, so $\mathrm{diag}(G)$ anywhere on the orbit follows from $\mathrm{diag}(G)$ at θ in closed form.

Lemma 2 (Hessian and Gauss–Newton diagonals coincide)**.** *Let the network consist of affine layers (dense or convolutional) and piecewise-linear activations and pooling (ReLU, max-pooling), with no*

normalization layers, and let the loss be $L(\theta) = \frac{1}{N}\sum_n \ell(f(x_n;\theta), y_n)$ *with* ℓ *twice differentiable in* f *(cross-entropy). Then at every* θ *at which the activation pattern is locally constant on the data, which holds for almost every* θ, $\operatorname{diag} H(\theta) = \operatorname{diag} G(\theta)$. *In particular* $\operatorname{tr} H = \operatorname{tr} G$ *and* $\sum_i H_{ii}^p = \sum_i G_{ii}^p$.

Proof. $H = G + \frac{1}{N}\sum_n \sum_k \frac{\partial \ell}{\partial f_k} \nabla_\theta^2 f_k(x_n;\theta)$. Each scalar parameter θ_i belongs to exactly one affine layer, in which it may appear at many positions (shared convolution weights); that layer's output is linear in θ_i, and no other layer depends on it. With all other parameters and the activation pattern (including max-pooling selections) fixed, the remaining layers act linearly, so f is affine in θ_i, $\partial^2 f_k / \partial \theta_i^2 = 0$, and the second term has a zero diagonal. □

The argument, and the lemma, fail when the output is not affine in a single parameter: smooth activations (GELU, SiLU, tanh), normalization layers, parameters shared across layers (tied weights), and multiplicative interactions such as gating or attention.

The residual term is not zero off the diagonal (weights of different layers multiply), so $H \neq G$ as matrices. Spectral quantities therefore differ, and we always compute $\lambda_{\max}$ from H (Hessian–vector products). Everything built from the diagonal, namely the trace, MS_p, the orbit excess and the decomposition below, is the same for H and G in the MLPs and the CNN of this paper; in particular MS_1 is the minimal Hessian trace over the orbit, as defined by Ibayashi et al. [18]. We write $\operatorname{tr} G$ for the exact value and "Hessian trace (Hutchinson)" for its stochastic estimate. In float64, $\operatorname{diag} H$ (from one Hessian–vector product per parameter) and $\operatorname{diag} G$ agree to 5×10^{-16} (relative) for an MLP and for a reduced-size version of the CNN (`scripts/test_hessian_diag.py`). Across the 913 loss-matched MLP checkpoints of the main, intervention and replication sweeps, the Hutchinson estimate of $\operatorname{tr} H$ agrees with the exact $\operatorname{tr} G$ within its sampling error (median ratio 0.96–1.02 per sweep and loss target; median relative standard error 2.4–5.8%). The lemma fails with BatchNorm, where a weight also enters the batch statistics: in the same check $\operatorname{diag} H$ and $\operatorname{diag} G$ differ by up to 94% of the largest entry. The BatchNorm experiments (§5.10) therefore use only Hessian-based measures, computed from Hessian–vector products, and no orbit decomposition.

Definition 1 (Minimum sharpness and orbit excess)**.** *Following Ibayashi et al. [18], and generalizing to powers* $p > 0$, $\mathrm{MS}_p(\theta) = \min_{\theta' \in \mathrm{orbit}(\theta)} \sum_i G_{ii}(\theta')^p$. *The minimum is attained whenever every hidden unit has at least one incoming and one outgoing parameter with* $G_{ii} > 0$*: the objective, written in the logarithms of the unit scales, then grows without bound in every direction (follow the unit with the largest or smallest log-scale to an edge that leaves or enters it). Units with zero curvature on both sides contribute nothing and do not affect the value; if a unit has zero curvature on one side only,* MS_p *is an infimum approached as that unit's scale goes to* 0 *or* ∞. MS_1 *is the minimal trace;* $\mathrm{MS}_{1/2}$ *is the minimum of the regularizer that Adam implicitly minimizes [29]. The* orbit excess *is* $E(\theta) = \operatorname{tr} G(\theta) / \mathrm{MS}_1(\theta) \geq 1$, *so that*

$$\operatorname{tr} G(\theta) = \underbrace{\mathrm{MS}_1(\theta)}_{\textit{orbit minimum}} \times \underbrace{E(\theta)}_{\textit{orbit excess}} . \tag{2}$$

The identity holds by definition; its content is that MS_1 is constant on the rescaling orbit, so it does not depend on where on the orbit the network sits. It is an invariant of the orbit, not of the input–output function: a network with redundant units (for example two units with identical incoming weights, between which the outgoing weights can be redistributed) can represent the same function on different orbits with different MS_1. We therefore call MS_1 the orbit-minimum term (the curvature left after minimizing over the rescaling orbit) and E the orbit excess; informally the first reflects the learned function and the second the coordinates, but only in this orbit-level sense. E measures how far the current coordinates are from the flattest representation. Comparing two

networks in $\log \operatorname{tr} G$ therefore splits *exactly* into a orbit-minimum term and a coordinate term. A weight rescaled by s contributes $G_{ii}^{p} s^{-2p}$, so the objective is a posynomial in the unit scales and is convex in their logarithms. Exact coordinate descent has a closed form: unit j sees $A_j c_j^{-2p} + B_j c_j^{2p}$, minimized at $c_j^{2p} = \sqrt{A_j / B_j}$, where A_j and B_j aggregate its incoming and outgoing curvature. At $p = 1$ the optimum is characterized by *curvature balance*, with equal incoming and outgoing curvature for every unit. This differs from the minimum-norm ("balanced") point, where incoming and outgoing *norms* are equal. We compute $\operatorname{diag}(G)$ exactly. For MLPs, one backward pass per class suffices, accumulating $\sum_n p_{nk} (\delta_{nk}^2)^\top (a_n^2)$ per layer; for CNNs we use per-sample gradients. After that, MS_p takes milliseconds.

4 Experimental protocol

Data and models. We use MNIST and CIFAR-10 with a class-balanced 10k-example training subset, the full test set, and per-channel input standardization. The models are an MLP in–128–64–10 (the architecture of the original study) and a small CNN (two 3×3 conv layers with 32 and 64 channels, each followed by 2×2 max-pooling, then FC 128 and FC 10). Neither has normalization layers. **Optimizers.** SGD, SGD with momentum 0.9 (SGD+M), SAM ($\rho = 0.05$, SGD+M base) and Adam (default βs, $\epsilon = 10^{-8}$). For each optimizer we use two base learning rates and batch sizes $\{32, 128, 512\}$ with square-root scaling $\eta = \eta_0 \sqrt{B/32}$, and three seeds, for 72 runs per dataset. A calibration sweep chose the base rates so that every configuration trains stably (Appendix D). **Loss matching.** Comparing optimizers after a fixed number of epochs conflates optimizer effects with how far training has progressed, and cross-entropy curvature shrinks with the loss. Each run is therefore measured when its full training loss first falls below each of $\{1, 0.3, 0.1, 0.03, 0.01\}$, and only checkpoints at the same target are compared. All 144 MLP runs reached every target. (One CIFAR-10 SGD+M run crossed 0.03 and 0.01 within the same epoch; its checkpoint is used for 0.01 only.) **Measures.** On a fixed subset of 4096 training examples we compute 17 measures (Appendix A). *Coordinate-dependent:* the original metric (relative isotropic Gaussian sharpness on the test set, $\sigma = 0.01$); isotropic average sharpness; the Hessian trace, both as a Hutchinson estimate [17] and exactly as $\operatorname{tr} G$ (the same quantity by Lemma 2); $\lambda_{\max}$ (Lanczos); worst-case SAM sharpness (10-step PGD); $\sum_i \sqrt{G_{ii}}$; and several of these at the minimum-norm point (*canonicalized*). *Rescaling-invariant:* MS_1, $\mathrm{MS}_{1/2}$, multiplicative (elementwise-relative) average sharpness, ASAM worst-case sharpness [26], and $\sum_i w_i^2 G_{ii}$. We also report filter-normalized average sharpness [28], which is often used as a scale-invariant measure. It is invariant to scaling a unit's incoming weights, but *not* to the compensating scaling of its outgoing weights, which changes the norms of the next layer's rows unevenly. A random rescaling with log-scale s.d. 1 changes it by a factor of 9.5 in our MLP, while the other perturbation measures above are unchanged to within 10^{-4} (`scripts/test_invariance.py`). We therefore treat it as *partially* invariant. **Statistic.** For a measure S we report $P(S_{\mathrm{Adam}} < S_X)$ over all pairs of Adam and X configurations sharing a seed and loss target (108 pairs). This is a common-language effect size, with 0.5 meaning no difference. For the trace we also report the exact split of $\log_2(\overline{\operatorname{tr} G}_{\mathrm{Adam}} / \overline{\operatorname{tr} G}_X)$ (geometric means) into function and coordinate terms via (2). 95% confidence intervals come from a bootstrap that resamples runs within each seed, independently for the two optimizers (2000 resamples). The resampling unit is the training run, so the dependence between the many pairs that share a run is accounted for. The intervals treat the six configurations per seed as exchangeable; they describe uncertainty over this learning-rate and batch-size grid, not over optimizers in general. The loss targets are measured on the same runs, so agreement across targets is consistency, not independent replication.

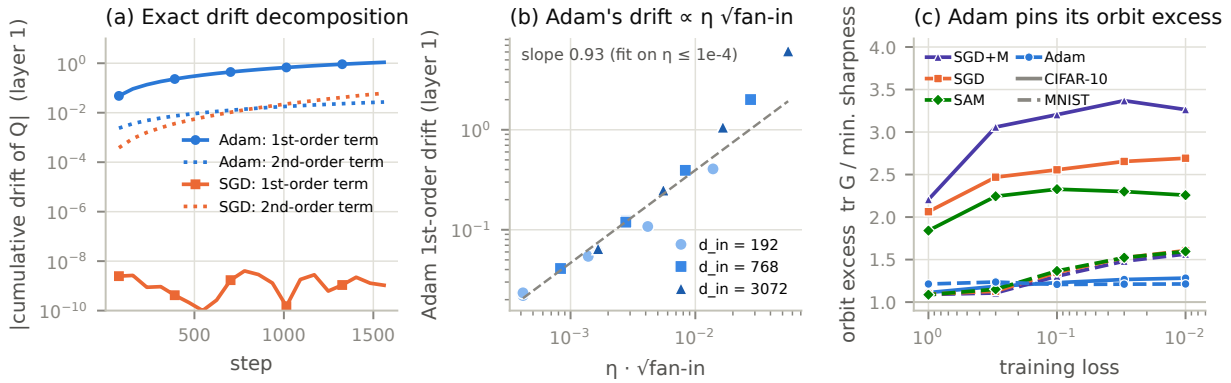


Figure 2: **(a)** Exact decomposition (1) of the cumulative drift of Q_j (mean over first-layer units; CIFAR-10 MLP, 32×32 inputs). SGD's first-order term is zero up to float32 round-off ($\sim 10^{-9}$), as Proposition 1a requires, while Adam's first-order term dominates its own second-order term. **(b)** Adam's first-order drift after 20 epochs, for four learning rates and three input resolutions (fan-in 192, 768 and 3072), plotted against $\eta\sqrt{\text{fan-in}}$. **(c)** Orbit excess $E = \operatorname{tr} G / \mathrm{MS}_1$ at matched training loss (median of 18 runs; solid: CIFAR-10, dashed: MNIST). Adam holds $E \approx 1.2$ on both datasets; the other optimizers' excess depends on dataset and training stage.

5 Results

5.1 Adam drifts along the orbit; SGD does not

We first isolate the mechanism. We log both terms of (1) at every step on the CIFAR-10 MLP, varying the learning rate ($3{\times}10^{-5}$–10^{-3} for Adam, 0.01–0.1 for SGD, 10^{-3}–10^{-2} for SGD+M) and the first layer's fan-in (inputs average-pooled to 8×8, 16×16 and 32×32), with two seeds per setting (66 runs). The decomposition closes to $\leq 1.3{\times}10^{-6}$ (float32 round-off) in every run.

SGD conserves; Adam drifts. SGD's first-order term is below $3{\times}10^{-7}$ in every run, zero up to round-off as Proposition 1a requires. Adam's is 0.02–6.2, five to seven orders of magnitude larger, and exceeds its own second-order term by 23–800× (Fig. 2a). SGD+M drifts only at second order, as Proposition 1b predicts. At equal effective learning rate $\eta/(1-\beta)$ its total drift matches SGD's (0.61 vs. 0.58 at $\eta_{\text{eff}} = 0.1$, 32×32).

Scaling law. A log-linear fit of Adam's first-layer first-order drift on $\log \eta$ and $\log \sqrt{d_{\text{in}}}$ gives exponents 0.91 and 0.94 for $\eta \leq 10^{-4}$, close to the $\eta\sqrt{d_{\text{in}}}$ predicted by Remark 1 (Fig. 2b). Over all learning rates the fit gives 1.07 and 1.38 ($R^2 = 0.97$). The steeper fan-in dependence at large η is consistent with the remark, since the ℓ_1 norms that drive the drift keep growing as training progresses further. The drift of the second layer, whose fan-in is fixed at 128, does not depend on input resolution. Seeds agree to within a few percent.

Where the drift leads. At initialization the CIFAR-10 MLP has orbit excess $E = 3.66$; MNIST has 1.79. During training, Adam moves to a median $E = 1.11$–1.28 on CIFAR-10 and 1.21–1.24 on MNIST, at every loss level (Fig. 2c). Across all 90 Adam checkpoints, E lies in $[1.08, 1.48]$ on CIFAR-10, against $[1.45, 4.89]$ for the other optimizers, and in $[1.12, 1.48]$ on MNIST. As the η-proportional drift predicts, the excess is slightly higher at Adam's smaller base learning rate (CIFAR-10 median 1.29–1.38 vs. 1.15–1.17), where there is less drift to carry the network away from its initial coordinates. SGD cannot move along the orbit at first order, and SGD+M and SAM move along it only weakly (Proposition 1b,d; in Table 5 their cumulative first-order terms

are 0.069 and 0.048 at training losses 1.10 and 0.90, against 0.255 for Adam at 1.13), so their excess follows from their nearly conserved Q_j combined with how the curvature redistributes during training; the second factor dominates, and SGD reaches a similar excess from very different starting points (§5.5). Their median excess is 1.8–3.4 on CIFAR-10 and drifts from 1.1 to 1.6 on MNIST. In parameter space, Adam's first-layer imbalance $|\log(\|a_j\|/\|c_j\|)|$ reaches 0.82–1.07 on CIFAR-10 (an incoming/outgoing norm ratio of about 2.9), compared with 0.30–0.46 for the other optimizers, which stay near the initial 0.35.

Not specific to Adam. Proposition 1c covers every diagonal preconditioner. Repeating the drift experiment with RMSprop and with plain sign descent (signSGD, which has neither momentum nor second-moment estimates) gives the same picture (Table 5). Both drift at first order, and with 32×32 inputs all three adaptive methods end at an orbit excess of 1.10–1.28, against 1.68–2.41 for SGD and SGD+M at comparable training losses. The common ingredient is that every coordinate takes a step of roughly the same size.

Where the drift leads: two phases. The data and the theory point to two distinct phases.

Noise-driven phase (slow). SGD is not standing still on the orbit. Its second-order term has expectation $\mathbb{E}D_j^{(2)} = \eta^2(\mathbb{E}\|g_{a_j}\|^2 - \mathbb{E}\|g_{c_j}\|^2)$. Near a minimum, minibatch gradients are dominated by noise and the Fisher matrix approximates the Gauss–Newton matrix, so this is approximately $\frac{\eta^2}{B}(A_j - B_j)$. Here $A_j = \sum_{i\in\mathrm{in}(j)} G_{ii}$ and $B_j = \sum_{k\in\mathrm{out}(j)} G_{kk}$ are the unit's incoming and outgoing curvature. A positive drift enlarges the incoming weights, which lowers A_j ($\propto c_j^{-2}$) and raises B_j ($\propto c_j^2$). The drift therefore pushes every unit toward *curvature balance* $A_j = B_j$, the stationarity condition of MS_1. This is the "law of balance" of Ziyin et al. [48, 49] written in curvature terms, and the orbit component of the $\operatorname{tr} H$-minimizing implicit bias of label-noise SGD [7, 10]; see also implicit gradient regularization [5].

The same argument can be applied to Adam. For a quasi-static diagonal preconditioner P, Adam's update is an SGD step in the coordinates $\theta' = P^{1/2}\theta$. Because P is diagonal it commutes with the rescaling, so Lemma 1 still holds in θ'. The noise covariance in these coordinates is $P^{-1/2}\Sigma P^{-1/2}$. With $p_i \approx \sqrt{\mathbb{E}g_i^2}$, noise balance then requires $\sum_{\mathrm{in}}\sqrt{\Sigma_{ii}} = \sum_{\mathrm{out}}\sqrt{\Sigma_{kk}}$, i.e. $\sum_{\mathrm{in}}\sqrt{G_{ii}} = \sum_{\mathrm{out}}\sqrt{G_{kk}}$. That is the stationarity condition of $\mathrm{MS}_{1/2}$, the orbit restriction of the $\operatorname{tr}\operatorname{diag}(H)^{1/2}$ regularizer that Li et al. [29] derive rigorously for Adam. The heuristic therefore predicts *different* long-run destinations: minimum trace for SGD and minimum $\sum_i \sqrt{G_{ii}}$ for Adam, both reached at $O(\eta^2)$ per step.

Signal-driven phase (fast). Early in training the gradient has a large mean, and Adam's first-order term (Proposition 1c) dominates. Empirically it takes Adam from an orbit excess of about 3.7 at initialization to ≈ 1.1 within about five epochs on CIFAR-10 (§5.5), i.e. close to the trace-minimizing point. We have only a heuristic for this destination. Adam's per-coordinate steps have roughly equal size η whatever the curvature, so if its update noise is roughly uncorrelated across coordinates with roughly equal variance r^2 per coordinate, the loss it injects, $\frac{1}{2}\operatorname{tr}(HC)$ for update-noise covariance C, is about $\frac{r^2}{2}\operatorname{tr} H$ in the current coordinates. Equal step sizes do not by themselves guarantee this covariance; it is an assumption of the heuristic, not something we measured. SGD has no first-order drift, so in this phase it stays wherever its initialization and its conserved Q_j put it.

Ordinary training, and every optimizer comparison in §5.2, lives in the regime where the fast phase has acted and the slow phase has not. §5.5 tests both phases directly.

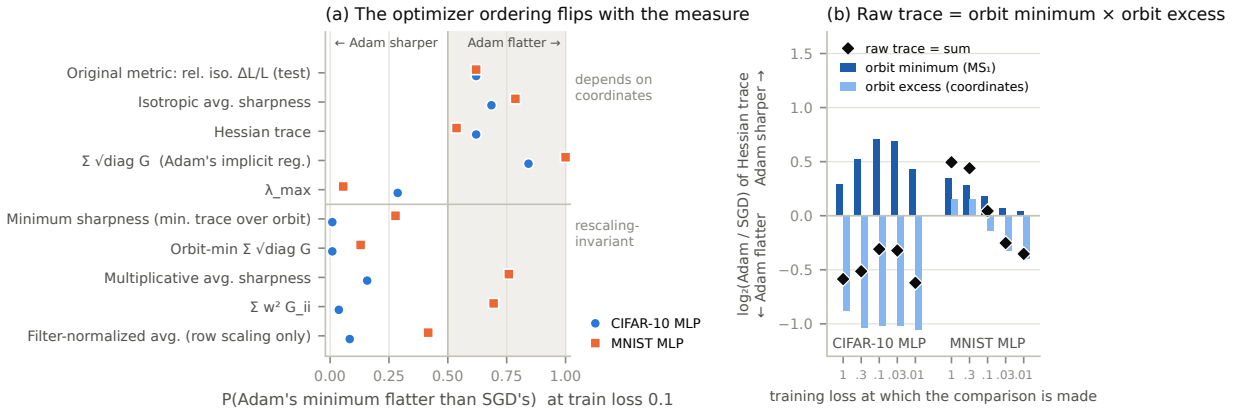


Figure 3: **(a)** Probability that Adam's minimum is rated flatter than SGD's, over all 108 loss-matched configuration pairs at training loss 0.1. **(b)** Exact split (2) of the Adam/SGD ratio of Hessian traces ($= \operatorname{tr} G$, Lemma 2) into a orbit-minimum term (minimum sharpness) and a coordinate term (orbit excess), at each matched training loss. On CIFAR-10, the coordinate term favours Adam by $\approx 2\times$ and outweighs the orbit-minimum term, which favours SGD. On MNIST, the coordinate term changes sign during training and flips the raw ordering, while the orbit-minimum term never favours Adam.

5.2 Consequence: the coordinate factor reverses optimizer rankings

CIFAR-10. Figure 3 and Table 2 summarize the audit. At every matched training loss, Adam is rated flatter than SGD by the original metric ($P = 0.62$–0.67), isotropic average sharpness (0.69–0.75), the Hessian trace (0.62–0.74) and, most strongly, $\sum_i \sqrt{G_{ii}}$ (0.84–1.00). Minimum sharpness reverses the conclusion ($P = 0.01$–0.22), as do $\mathrm{MS}_{1/2}$ (0.01–0.19), filter-normalized sharpness (0.06–0.14), multiplicative sharpness (0.16–0.31) and $\sum w_i^2 G_{ii}$ (0.04–0.24). With 18 runs per optimizer the raw verdicts carry substantial uncertainty (e.g. $\operatorname{tr} G$ at loss 0.1: $P = 0.64$, 95% CI $[0.44, 0.83]$), whereas the invariant verdicts are tight (MS_1: 0.01, $[0.00, 0.06]$). The quantity that matters, the disagreement between them, is well determined: $P_{\operatorname{tr} G} - P_{\mathrm{MS}_1} = +0.69$ $[+0.52, +0.83]$, $+0.63$ $[+0.43, +0.81]$ and $+0.50$ $[+0.34, +0.67]$ at losses 0.3, 0.1 and 0.01. The exact split explains this. In bits, $\log_2(\operatorname{tr} G_{\mathrm{Adam}} / \operatorname{tr} G_{\mathrm{SGD}})$ decomposes into a orbit-minimum term of $+0.52$ $[+0.36, +0.68]$, $+0.71$ $[+0.53, +0.87]$ and $+0.43$ $[+0.16, +0.71]$ (Adam sharper), and a coordinate term of -1.03 $[-1.17, -0.91]$, -1.02 $[-1.16, -0.87]$ and -1.05 $[-1.22, -0.89]$, a factor of almost exactly 2 in Adam's favour at every loss level. Against SGD+M the coordinate term is larger still (-0.94 to -1.33 bits). Against SAM, the one optimizer designed to seek flat minima, the raw trace is inconclusive ($P = 0.44$–0.60) because a large orbit-minimum term ($+0.69$ to $+1.00$ bits, SAM flatter) is cancelled by the coordinate term (-0.69 to -0.92). Invariant measures identify SAM as clearly flatter ($P = 0.00$–0.23). A measure that fails this sanity check should not be used to rank optimizers.

Robustness to the learning-rate pairing. Sharpness depends strongly on the learning rate, so we also split the comparison by the pair of base learning rates (Table 6, pooled over batch sizes, seeds and losses 0.1–0.01). On CIFAR-10 the raw verdict is set by the pairing: P for $\operatorname{tr} G$ is 1.00, 0.06, 1.00 and 0.60 in the four pairings, and the original metric ranges from 0.00 to 1.00. Minimum sharpness rates Adam sharper in all four pairings ($P \le 0.27$), and so does filter-normalized sharpness ($P \le 0.09$). On MNIST both raw and invariant verdicts depend on the pairing (MS_1: 0.00–0.88), consistent with the small functional differences there.

MNIST. Here the initial excess is smaller and the picture shifts during training (Fig. 3b). Early on (training loss 1 to 0.3), SGD's excess is *below* Adam's. The coordinate term therefore slightly favours SGD ($+0.15$ $[+0.12, +0.19]$ bits at loss 0.3), and raw measures rate Adam sharper ($P = 0.06$ for the trace). As SGD's excess grows, the coordinate term turns in Adam's favour (-0.40 $[-0.41, -0.38]$ bits at loss 0.01) and the raw trace ordering flips ($P = 0.80$, $[0.63, 0.94]$). The original metric goes from 0.27 to 0.97. The orbit-minimum term never favours Adam: it shrinks from $+0.34$ to $+0.04$ bits (the latter indistinguishable from zero, $[-0.13, +0.22]$), and minimum sharpness gives $P = 0.09$, 0.28 and 0.44 at losses 0.3, 0.1 and 0.01. On MNIST, then, whether "Adam is flatter" depends on *when* one measures, and the dependence comes almost entirely from the coordinate factor.

Invariant measures need not agree with each other. Invariance removes the coordinate artifact, but it does not make different notions of flatness coincide. On CIFAR-10 all invariant measures agree, with one exception: large-radius ASAM at the earliest target ($P = 0.71$ at loss 0.3, then 0.32 and 0.28). On MNIST, the elementwise-relative family (multiplicative sharpness, ASAM, $\sum_i w_i^2 G_{ii}$) rates Adam flatter late in training ($P = 0.73$–0.87 at loss 0.01), while minimum sharpness and filter-normalized sharpness do not (0.44 and 0.20). The relative measures weight each weight's curvature by its squared magnitude, which is itself affected by Adam's sign-like updates. We therefore make no universal claim that Adam's minima *are* sharper. Our claim is narrower and robust: the raw-coordinate ordering is biased by an optimizer-dependent factor that is as large as the effects being measured.

$\lambda_{\max}$ and canonicalization. The top eigenvalue rates Adam sharper even in raw coordinates ($P = 0.29$–0.45 on CIFAR-10, 0.00–0.32 on MNIST), in line with prior reports and with edge-of-stability dynamics. It is dominated by a few outlier directions, while the drift mainly reshapes the bulk of the diagonal that trace-type and isotropic measures integrate. Evaluating the trace at each network's minimum-norm representative is itself invariant to the rescalings, yet it disagrees with MS_1 ($P = 0.31$–0.56 for the trace on CIFAR-10, against 0.01–0.22). Both are orbit invariants; they evaluate the curvature at different canonical points of the same orbit (norm balance versus curvature balance). Orbit invariance alone therefore does not settle the comparison, and the choice of canonical point matters. Adam's drift moves toward *curvature* balance, the stationarity condition of MS_1, not toward norm balance.

5.3 Causal test: removing Adam's drift

If the coordinate factor is caused by Adam's drift, removing the drift should change raw measures and leave invariant ones alone. *Adam-Q* runs Adam and, after every step, applies the function-preserving rescaling that restores each unit's Q_j to its initial value. For a single unit this requires solving $c^2\|a_j\|^2 - \|c_j\|^2/c^2 = Q_j^{(0)}$ for $c > 0$, which has a closed form. Because rescaling a unit also changes the incoming weights of the next layer, we apply these per-unit solutions layer by layer in two sweeps. The restoration is therefore approximate: in a check over 300 Adam steps on the CIFAR-10 MLP (inspected at steps 1, 100 and 300), the remaining deviation of Q_j from $Q_j^{(0)}$ was at most 7×10^{-4} of $\max_j |Q_j^{(0)}|$ in the first layer and 3×10^{-7} in the second, did not accumulate, and the network output changed by at most 4×10^{-6} per restoration. Adam's moment estimates are transformed covariantly: for a weight scaled by s, $m \mapsto m/s$ and $v \mapsto v/s^2$. This keeps Adam's state consistent with the new coordinates but does not make the dynamics rescaling-equivariant. Mapped back to the old coordinates, the next step becomes $-(\eta/s)\,\hat{m}/(\sqrt{\hat{v}} + s\epsilon)$ instead of the step $-\eta\,\hat{m}/(\sqrt{\hat{v}} + \epsilon)$ Adam would have taken there, so even with $\epsilon = 0$ it differs by the factor $1/s$, and the fixed $\epsilon = 10^{-8}$

Table 2: P(Adam rated flatter than X) at matched training loss L (108 configuration pairs per cell). Values below 0.5 mean Adam is rated sharper.

		CIFAR-10, vs SGD			CIFAR-10, vs SAM			MNIST, vs SGD		
Measure	type	L=0.3	0.1	0.01	0.3	0.1	0.01	0.3	0.1	0.01
Original metric (rel. iso., test)	raw	0.66	0.62	0.67	0.61	0.58	0.56	0.27	0.62	0.97
Isotropic average	raw	0.74	0.69	0.75	0.67	0.65	0.58	0.41	0.79	0.90
Hessian trace (Hutchinson)	raw	0.74	0.62	0.74	0.58	0.52	0.46	0.06	0.54	0.83
$\sum_i \sqrt{G_{ii}}$	raw	1.00	0.84	0.92	0.89	0.74	0.71	1.00	1.00	1.00
$\lambda_{\max}$	raw	0.37	0.29	0.45	0.06	0.04	0.07	0.00	0.06	0.32
$\operatorname{tr} G$ at min-norm point	canon.	0.56	0.31	0.53	0.32	0.19	0.20	0.61	0.77	0.81
Minimum sharpness MS_1	inv.	0.06	0.01	0.22	0.00	0.00	0.06	0.09	0.28	0.44
$\mathrm{MS}_{1/2}$	inv.	0.03	0.01	0.19	0.00	0.00	0.05	0.00	0.13	0.31
Filter-normalized average	part.	0.14	0.08	0.06	0.06	0.02	0.05	0.47	0.42	0.20
Multiplicative average	inv.	0.31	0.16	0.25	0.23	0.10	0.14	0.57	0.76	0.73
$\sum_i w_i^2 G_{ii}$	inv.	0.18	0.04	0.24	0.00	0.00	0.11	0.54	0.69	0.84
ASAM worst-case (ρ=0.5)	inv.	0.71	0.32	0.28	0.07	0.03	0.15	0.45	0.68	0.87

adds a further, here negligible, difference. (Plain SGD is not equivariant to these rescalings either: in the same comparison its step changes by $1/s^2$. What distinguishes SGD is that its first-order drift along the orbit vanishes, Proposition 1.) Every function-preserving rescaling therefore changes the subsequent trajectory, which is why Adam-Q is compared with Adam only at matched training loss. Since the rescaling preserves the function, Adam-Q optimizes the same loss and differs from Adam only because the optimizer's steps depend on where on the orbit the network sits. We rerun the 18 Adam configurations of each main sweep with Adam-Q and compare checkpoints of the same configuration whose training losses agree to within 25% (73 of 78 pairs on CIFAR-10).

In 12 of the 18 CIFAR-10 configurations (base learning rate 10^{-4}, and all runs at batch size 512) Adam-Q trains like Adam, reaching every loss target in the same or slightly fewer epochs. In the six noisiest configurations (base rate 3×10^{-4} at batch sizes 32 and 128), however, Adam-Q stalls. It reaches loss 0.1 two to three times later than Adam and never gets below ≈ 0.05 within 300 epochs, whereas Adam reaches 0.01 by epoch ≈ 55. On MNIST all 18 Adam-Q runs train normally. So the drift is not merely a measurement nuisance: in noisy regimes it appears to help Adam train. Three controls, each run with two seeds at the three base settings, narrow this down. First, restoring Q_j only every 50 steps instead of every step still stalls at batch sizes 32 and 128, so the stall is not caused by intervening at every step. Second, restoring Q_j in only one hidden layer trains normally. Freezing just the first layer, which has fan-in 3072 and the largest drift, reaches loss 0.01 in 57–78 epochs; freezing just the second takes 52–62, against 52–67 for Adam. The obvious hypothesis, that Adam must inflate its high-fan-in weights to shrink their relative step $\eta/|w_i|$, is therefore wrong: the stall appears only when both layers are prevented from drifting. Third, restoring Q_j *without* transforming Adam's moment estimates stalls at every batch size, including 512. That version is a harsher intervention rather than a cleaner control, which is why the covariant transformation is part of Adam-Q.

Where Adam is held, not whether it is held, causes the stall. A fourth control separates the intervention from the orbit position. Every 100 steps we move Adam, with its moment estimates transformed covariantly, to a chosen point of its current orbit and let it drift freely in between: either to the minimum-trace point, or to the minimum-norm point, which is where Adam-Q ends up (its imbalance falls to 0.02). We use the same three settings and two seeds (12 runs). Held

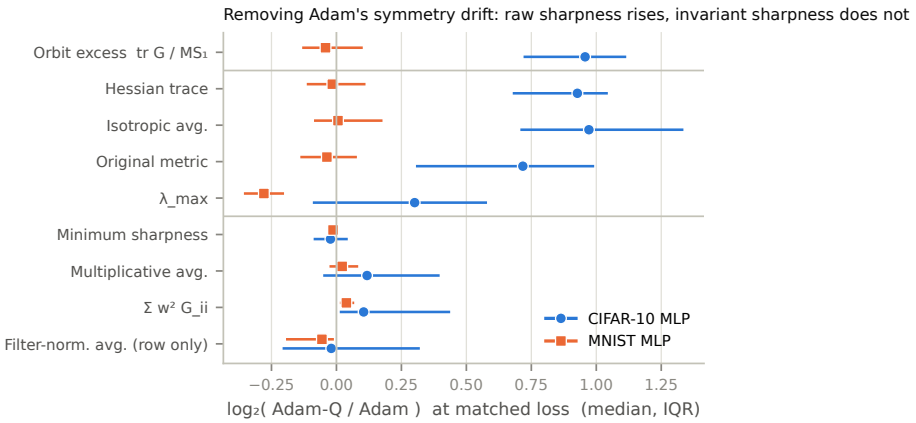


Figure 4: **Causal test.** Ratio of each measure for Adam-Q (Adam with its symmetry drift removed after every step) to Adam, for the same learning rate, batch size and seed at matched training loss (median and IQR over loss-matched checkpoint pairs). On CIFAR-10, removing the drift doubles the orbit excess and, with it, every raw trace-type measure. On MNIST, where the conserved coordinates are already close to the orbit minimum, the excess barely moves and neither do the raw measures. Invariant measures are unchanged in both cases.

at the minimum-trace point (orbit excess 1.00–1.01 at every checkpoint), Adam trains as fast as unconstrained Adam: all six runs reach loss 0.01, in 51–61 epochs against 52–67. Held at the minimum-norm point (excess 2.1–2.8 from loss 0.3 on), all four runs at batch sizes 32 and 128 stall like Adam-Q, at a final loss of 0.17–0.30 after 300 epochs (Adam-Q: 0.18–0.30), while the two runs at batch size 512 still reach 0.01. The intervention itself is therefore harmless, and what slows Adam is being kept at coordinates with a large orbit excess. This is what the heuristic of §5.1 predicts under its assumption of roughly uncorrelated, equal-variance update noise: the loss injected by Adam's update noise then scales with the trace in its *current* coordinates, about 2× larger at the minimum-norm point than near the minimum-trace point, and the difference matters only when the gradient noise is large. Adam's drift toward the minimum-trace point thus lowers its own noise floor; it helps Adam train as well as making its minima look flat. All comparisons below use only checkpoint pairs that are genuinely loss-matched. Across them, test accuracy at matched loss differs by a median factor of 0.995. Adam-Q's coordinates stay SGD-like: first-layer imbalance falls from 0.89 to 0.13 and the orbit excess rises from 1.18 to 2.35. Figure 4 shows the consequence on CIFAR-10. Every raw trace-type measure roughly doubles: $\operatorname{tr} G$ by a median factor of 1.91 (IQR 1.63–2.04), the isotropic average by 1.96 and the original metric by 1.64. The invariant measures are essentially unchanged: MS_1 0.98 (IQR 0.94–1.03), $\mathrm{MS}_{1/2}$ 0.99, multiplicative 1.09, and the partially invariant filter-normalized measure 0.99. $\lambda_{\max}$ moves less (1.23, IQR 0.94–1.49), consistent with its weaker dependence on the coordinate factor (§5.2). Ranked against SGD, Adam-Q is no longer rated flatter: $P(\text{flatter than SGD})$ by $\operatorname{tr} G$ drops from 0.64–0.75 (Adam) to 0.21–0.35 (Adam-Q), and by the original metric from 0.62–0.67 to 0.12–0.34. Minimum sharpness gives the same verdict for both (0.06/0.01/0.22 vs. 0.00/0.02/0.21 at losses 0.3/0.1/0.01; at 0.01 only the 12 Adam-Q configurations that reach it are included). The one deviation is at the lowest loss target (0.01), where the median MS_1 ratio is 0.88. This is not a loss-matching artifact: the achieved loss divided by the target has a median of 0.94–0.98 for both Adam and Adam-Q at every target, and tightening the matching band from 25% to 10% and 5% (67 and 48 pairs) leaves the medians essentially unchanged ($\operatorname{tr} G$ ratio 1.91, 1.89, 1.87; MS_1 ratio 0.98, 0.99, 0.98; at loss 0.01 still 0.88). At the lowest loss, Adam-Q's networks

are slightly flatter than Adam's even in orbit-minimized sharpness. Six Adam-Q configurations never reach loss 0.01, so comparisons there involve a selected subset. Restricting Adam to the same 12 configurations does not weaken the main result: $P(\text{Adam flatter than SGD})$ at loss 0.01 becomes 0.62 (from 0.72) by $\operatorname{tr} G$ and 0.01 (from 0.22) by MS_1.

On MNIST the intervention changes little, which is what the decomposition predicts. MNIST's conserved coordinates happen to lie close to the orbit minimum (initial excess 1.79 vs. 3.66 on CIFAR-10), so Adam-Q's excess (1.11–1.36 across loss targets, median 1.17) stays near Adam's (1.21). The raw measures accordingly move by at most about 13% (median $\operatorname{tr} G$ ratio 0.88–1.07 across targets), and the invariant measures are again unchanged (MS_1 0.99, IQR 0.98–1.00; $\mathrm{MS}_{1/2}$ 0.99). Together the two datasets give a dose–response: Adam-Q moves raw sharpness in proportion to how far it moves the coordinates ($\times 1.9$ when the excess doubles on CIFAR-10, about $\pm 10\%$ when it barely changes on MNIST), and moves minimum sharpness in neither case. Only $\lambda_{\max}$ behaves differently on MNIST ($\times 0.82$), a reminder that it is not rescaling-invariant either.

What the intervention does and does not show. Each Adam-Q rescaling preserves the network output to float32 precision (3×10^{-6}, Appendix A), uses the same loss, minibatches, learning rate and schedule as Adam, and transforms the optimizer state covariantly. It does *not* leave the later trajectory unchanged: because Adam is not rescaling-equivariant, every subsequent step differs, which is how the stall arises. We therefore compare only checkpoint pairs that are matched in training loss, never runs at equal epochs. Part of the result is also arithmetic: since $\operatorname{tr} G = \mathrm{MS}_1 \times E$ exactly, doubling E doubles the trace unless MS_1 changes. The empirical content is that it does not change ($\times 0.98$): at matched loss, the drift changes where Adam sits on the orbit but not, measurably, the orbit-minimized sharpness or test accuracy of what it finds. Removing Adam's flatness advantage over SGD is the consequence of that. The intervention establishes that Adam's drift causes the coordinate factor in these MLPs; it does not, on its own, establish anything about why Adam generalizes as it does.

5.4 Are Adam's coordinates good coordinates for SGD?

If Adam ends near the trace-minimizing point and SGD does not, would SGD benefit from being moved there? Symmetry teleportation makes this cheap to test [45, 46], since our target has a closed form. Every 100 steps we map SGD (or SGD+M) to the minimum-sharpness point of its current orbit, computed from $\operatorname{diag}(G)$ on 1024 training examples, and transform the momentum buffer covariantly. We use four learning rates per optimizer and two seeds on the CIFAR-10 MLP.

It does not help; it destabilizes training. Ten of the 16 teleported runs diverged, against none of the 16 plain runs. At $\eta = 0.12$, for instance, plain SGD reaches loss 0.01 in 52–53 epochs, while both teleported runs diverge. The two teleported runs that did reach loss 0.01 generalized worse (test accuracy 0.392–0.396 vs. 0.433–0.441).

The reason is the top eigenvalue. Logging $\lambda_{\max}$ around every teleport of an $\eta = 0.12$ run, each teleport *raised* it by 3–20%, leaving it within 5% of, or above, SGD's stability threshold $2/\eta = 16.7$ in 8 of 11 teleports. SGD on its own keeps $\lambda_{\max}$ just below $2/\eta$, the self-stabilization of gradient descent at the edge of stability [8, 11], and the teleports keep knocking it over. The teleport scales were mild (0.95–2.5), so this is not an artifact of extreme rescalings. Measured on the converged networks, the same map *lowers* $\lambda_{\max}$ (e.g. $15.9 \to 11.0$), so the trace and the top eigenvalue are minimized at different points of the orbit, and which point is "better" depends on what the optimizer needs. Adam's preferred coordinates are good for Adam, not for SGD.

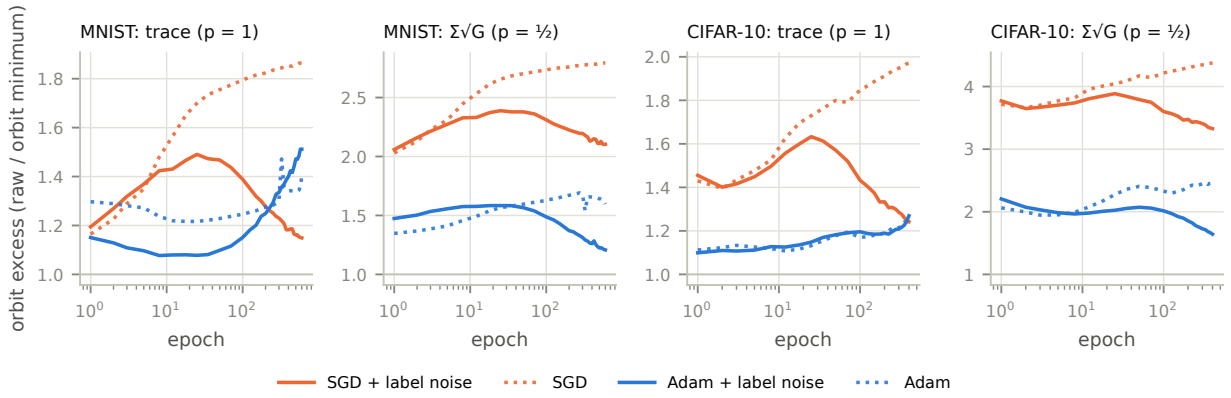


Figure 5: **Where each optimizer drifts on long horizons.** Orbit excess for the trace ($p = 1$) and for $\sum_i \sqrt{G_{ii}}$ ($p = \frac{1}{2}$) during 600 (MNIST) and 400 (CIFAR-10) epochs, with and without label noise (SGD $\eta = 0.12$, Adam $\eta = 6 \times 10^{-4}$, batch size 128; mean of two seeds; log-scaled epoch axis). Without label noise SGD's coordinates nearly stop moving once the data are fit. With label noise SGD moves toward the trace-minimizing point, and Adam moves toward the $\sum_i \sqrt{G_{ii}}$-minimizing point while its trace excess grows: the destinations predicted for label-noise SGD [7, 10] and for Adam [29].

5.5 Long horizons: the orbit as a testbed for implicit sharpness regularization

The two-phase picture makes predictions that ordinary training cannot test, because the slow phase needs persistent gradient noise and long horizons. Implicit-regularization theory supplies the standard device for this: *label noise*, in which every label is replaced by a uniformly random class with probability 0.2, freshly at every step [7, 10]. We train the MLPs for 600 (MNIST) and 400 (CIFAR-10) epochs with SGD ($\eta \in \{0.06, 0.12\}$) and Adam ($\eta \in \{2, 6\} \times 10^{-4}$), with and without label noise, two seeds each (32 runs), and track the orbit excess for $p = 1$ and $p = \frac{1}{2}$ (Fig. 5). Both targets are computable exactly. To our knowledge, the predictions of Damian et al. [10] and Li et al. [29] had so far been checked only in diagonal linear networks and matrix factorization. The rescaling orbit gives a direction in a nonlinear network along which the predicted destination is known in closed form.

Fast phase. From an orbit excess of about 3.7 at initialization, Adam reaches 1.09–1.13 by epoch 5 on CIFAR-10 in all four Adam settings, while SGD is at 1.45–1.63.

Without noise, SGD freezes. SGD's excess rises while the data are being fit and then stops changing once gradients vanish (1.78–1.86 on MNIST, 1.97–2.67 on CIFAR-10).

With label noise, SGD moves toward minimum trace. After the initial transient its trace excess falls steadily: on CIFAR-10 from 2.05 (epoch 25) to 1.63 at $\eta = 0.06$, and from 1.63 to 1.24 at $\eta = 0.12$; on MNIST from 1.65 (epoch 100) to 1.35, and from 1.39 to 1.15. The movement is faster at the larger learning rate, as a second-order drift should be. Its per-unit curvature balance at $p = 1$ shrinks markedly (CIFAR-10, $\eta = 0.12$: mean $|\log A_j/B_j|$ from 1.37 to 0.80), while balance at $p = \frac{1}{2}$ barely changes (2.31 to 2.04).

With label noise, Adam moves toward minimum $\sum_i \sqrt{G_{ii}}$. At $\eta = 6 \times 10^{-4}$, Adam's $p = \frac{1}{2}$ excess falls (MNIST 1.56 $\to$ 1.21, CIFAR-10 1.99 $\to$ 1.64) while its trace excess *rises* (MNIST 1.10 $\to$ 1.51, CIFAR-10 1.11 $\to$ 1.27).

A per-run test. For each run we compare the change in log excess from epoch 100 to the end at $p = \frac{1}{2}$ with that at $p = 1$. In all 8 label-noise SGD runs the trace excess falls by more (mean

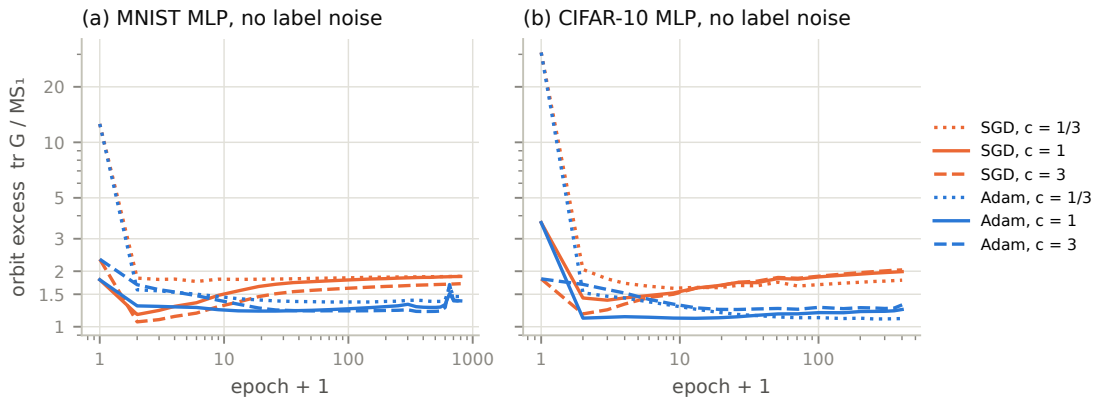


Figure 6: **Coordinate memory.** The same initial function started at three points of its orbit (first-layer weights $\times c$, second layer $\div c$), trained without label noise (mean of two seeds). The initial spread in orbit excess disappears within one epoch for both optimizers, but from every starting point Adam ends below SGD.

difference $+0.090$), and in all 8 label-noise Adam runs the $\sum\sqrt{G}$ excess falls by more (mean -0.219). These 8 runs per optimizer come from only 4 independent initializations; the replication and the block-level test below address this. Each optimizer thus drifts toward the orbit point favoured by its own implicit regularizer. The ordinary-training regime of §5.2 corresponds to the end of the fast phase, before this slow drift has had time to act.

Does the starting orbit point matter? Our account of SGD's excess leans on its conservation law, so we tested it directly. We start the *same initial function* at three points of its orbit, multiplying the first layer's incoming weights and biases by $c \in \{\frac{1}{3}, 1, 3\}$ and dividing the second layer's weights by c, and train with SGD and Adam, with and without label noise, for 800 (MNIST) and 400 (CIFAR-10) epochs (same learning rates as above, two seeds; 48 runs). The initial excess differs by a factor of 7.0 (MNIST) and 16.9 (CIFAR-10) across c (Fig. 6).[1]

The prediction that SGD would remember its starting point in the orbit excess fails, and the reason is informative. For every optimizer, the spread across c collapses to 1.3–1.9$\times$ within the first epoch and ends at 1.03–1.19$\times$ (one exception: 1.38$\times$ for Adam with label noise on CIFAR-10). The curvature redistribution of early training, not the starting coordinates, sets the excess. SGD does remember its starting point in *parameter* space: without label noise it keeps 90–99% of the initial separation in mean first-layer Q_j between $c = 3$ and $c = \frac{1}{3}$, and with label noise 79–83%. Adam does not erase that separation either. It enlarges it, to 1.5–2.0$\times$ in seven of eight runs (and 0.97$\times$ in one), consistent with a drift proportional to the units' ℓ_1 norms (Remark 1). What does not depend on the starting point is the gap between the optimizers. Without label noise Adam ends at an excess of 1.29–1.51 on MNIST and 1.09–1.35 on CIFAR-10 from every starting point, against 1.71–1.89 and 1.78–2.04 for SGD. Starting SGD at a low excess does not keep it there (CIFAR-10, $c = 3$: $1.82 \to 2.03$). Adam's low excess is therefore not an accident of the default initialization, and SGD's higher excess is where its own dynamics lead from any of these starting points.

The slow phase replicates across starting points. In the per-run test above (epoch 100 to the end), SGD's trace excess falls by more than its $\sum_i \sqrt{G_{ii}}$ excess in all 12 label-noise runs, and Adam's

[1]The CIFAR-10 runs of this experiment were done on a machine that could not reach the original CIFAR-10 host. We used a lossless PNG copy, whose ordering differs, so their 10k training subset differs from that of the other CIFAR-10 experiments.

$\sum_i \sqrt{G_{ii}}$ excess falls by more in all 12. These runs share initializations with each other and with the runs above, so they are not independent confirmations; the block-level test below accounts for this.

Independent replication and a frozen-curvature test. The runs above are not independent draws: runs that share a dataset and a seed start from the same network (the two learning rates here, the three orbit starts below), and the label-noise runs of the coordinate-memory experiment at $c = 1$ repeat settings of this section. We therefore added six seeds (2–7) per dataset at the learning rates the two experiments share (SGD $\eta = 0.12$, Adam 6×10^{-4}; 24 runs; CIFAR-10 runs use the mirrored copy of the data) and count independent initializations (dataset $\times$ seed) rather than runs. Across all three sets, 32 label-noise runs per optimizer come from 16 independent initializations. The per-run statistic has the predicted sign in all 32 runs of each optimizer, and the block means in all 16 blocks (sign test $p = 3 \times 10^{-5}$ each; mean block statistic $+0.073$ for SGD and -0.363 for Adam). The orbit excess changes both when the parameters move along the orbit and when the curvature $\mathrm{diag}(G)$ itself changes. To separate the two, in the 24 new runs we hold $\mathrm{diag}(G)$ at each snapshot fixed (snapshots every 25 epochs from epoch 100), move the parameters of that snapshot along their own orbit until every Q_j equals its value at the next snapshot (residual below 10^{-6}), and record how the frozen objectives $\sum_i G_{ii}$ and $\sum_i \sqrt{G_{ii}}$ change. Summed over a run, SGD's orbit motion lowers the frozen trace by more than the frozen $\sum_i \sqrt{G_{ii}}$ (mean change in log -0.46 vs. -0.35) in 12 of 12 runs, and Adam's raises the frozen trace while lowering the frozen $\sum_i \sqrt{G_{ii}}$ ($+0.16$ vs. -0.46) in 12 of 12 (sign test $p = 0.0005$ each). The motion along the orbit itself, not only the evolving curvature, is thus directed toward each optimizer's predicted point. This identifies the orbit component of the motion with the change in Q_j, which is one natural definition but not the only one.

Per-unit drift direction. Between consecutive snapshots we correlate each hidden unit's change in Q_j with its curvature imbalance $\log(A_j/B_j)$ at $p = 1$ and at $p = \frac{1}{2}$ (Spearman over units, averaged over snapshot pairs). Drift toward balance at power p predicts a positive correlation at that p. Adam's per-unit drift matches $p = \frac{1}{2}$ better than $p = 1$ in every run, window and setting (24 of 24 runs, early and late). With label noise, the late-phase correlation is $+0.49$ at $p = \frac{1}{2}$ against -0.06 at $p = 1$ on MNIST, and $+0.18$ against -0.16 on CIFAR-10. Without label noise Adam's units drift *away* from trace balance (-0.25 and -0.56; negative in 12 of 12 runs), as expected if the first-order sign-descent term rather than curvature dominates. For SGD the unit-level picture is mixed. On MNIST with label noise the drift matches $p = 1$ better ($+0.39$ vs. $+0.26$, 6 of 6 runs). On CIFAR-10 it correlates positively with both but more strongly with $p = \frac{1}{2}$ ($+0.60$ vs. $+0.76$, 0 of 6 runs favour $p = 1$), although at the orbit level the same runs move toward minimum trace. The Fisher $\approx$ Gauss–Newton step of our heuristic is the likely weak point here. We therefore regard the orbit-level destinations, which hold in all 64 label-noise runs (16 independent initializations per optimizer) and survive the frozen-curvature test, as established, and the unit-level mechanism for SGD as only partly confirmed.

5.6 Consequences for generalization prediction

This analysis is exploratory (the choice of target below was made after seeing the data). If the coordinate factor is a *cross-optimizer* bias, it should hurt generalization prediction mainly when models from different optimizers are ranked together. This is what we find (Fig. 7). Within one optimizer, raw $\mathrm{tr}\, G$ and MS_1 predict the accuracy gap equally well (mean Kendall τ at loss 0.01: 0.55 vs. 0.59 on CIFAR-10, 0.54 vs. 0.54 on MNIST). Pooled across optimizers, $\mathrm{tr}\, G$ drops to 0.38 (CIFAR-10) and 0.47 (MNIST), while MS_1 stays at 0.61 and 0.62. The gain from removing the coordinate factor is larger on CIFAR-10 ($+0.21$ to $+0.23$ across loss targets) than on MNIST ($+0.01$ to $+0.15$), matching the smaller coordinate term there. A bootstrap over runs (stratified by optimizer, 2000 resamples; it treats runs as independent and so does not model the dependence between runs

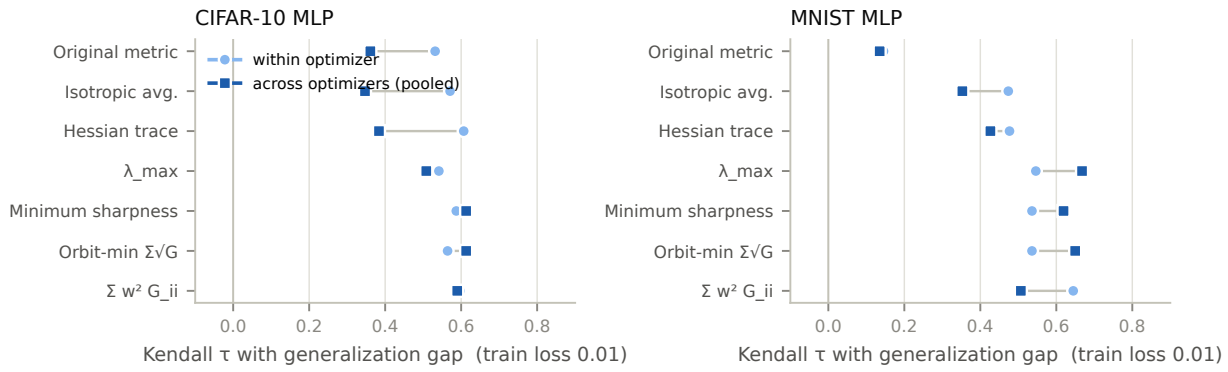


Figure 7: Kendall τ between each measure and the generalization gap (train minus test accuracy) at training loss 0.01, computed within each optimizer (averaged over the four optimizers) and pooled across optimizers. Raw trace-type measures lose predictive power when models from different optimizers are ranked together; minimum sharpness does not.

that share a seed, which three seeds are too few to estimate) gives a pooled improvement from $\operatorname{tr} G$ to MS_1 at loss 0.01 of $+0.24$, 95% CI $[+0.14, +0.33]$, on CIFAR-10 and $+0.15$ $[+0.09, +0.20]$ on MNIST. The original metric is the weakest cross-optimizer predictor (0.36 on CIFAR-10, 0.14 on MNIST). Not every raw measure behaves like the trace: $\lambda_{\max}$ and worst-case SAM sharpness, which depend less on the coordinate factor, reach a pooled τ of 0.49–0.61 on CIFAR-10 without any correction. We use the accuracy gap rather than the test loss as the target. This choice was made after seeing the test-loss correlations: at matched training loss in this memorization regime, the test loss is dominated by over-confident errors, and on CIFAR-10 it is negatively correlated with most measures (pooled τ between -0.46 and $+0.07$ at loss 0.01 for all 17; `scripts/gen_correlation.py` prints both tables). These are rank correlations over the configurations of two MLP sweeps. They show that the coordinate factor degrades cross-optimizer comparisons here; they do not show that invariant sharpness is a better predictor of generalization in general, or that sharpness causes generalization.

5.7 Revisiting the original study

Table 3 re-runs the original fixed-epoch protocol with five seeds. The original result replicates: the relative perturbation metric is about 2.9× lower for Adam on both datasets. The two datasets fail for different reasons, however. On **CIFAR-10**, both optimizers are far from convergence at similar training losses (1.29 vs. 1.42). The 2.5× gap in Hessian trace is almost entirely the coordinate factor ($E = 1.23$ vs. 2.85, i.e. 2.3×), the minimum sharpness is nearly equal (320 vs. 353), and the invariant multiplicative measure and the partially invariant filter-normalized measure rate Adam about 2.5× *sharper*. On **MNIST**, after 20 epochs Adam has reached a 17× lower training loss than SGD (0.008 vs. 0.136), and every measure, invariant or not, is correspondingly lower. That is a loss-level confound of fixed-epoch comparisons, not a property of the optimizer, and matching the loss removes it (§5.2). One further detail of the original study may also have contributed. Its single Adam run on CIFAR-10 reported only 33.7% test accuracy, against $49.6 \pm 0.5\%$ in our five-seed PyTorch replication. The difference may come from framework defaults (Keras uses Glorot initialization and $\epsilon = 10^{-7}$) or from an unstable run. In our pilot runs with raw, uncentred inputs, Adam at $\eta = 10^{-3}$ could oscillate around a training loss of ≈ 1. Either way, a high final loss lowers the relative metric $\Delta L/L$ mechanically.

Table 3: The original protocol, re-run with five seeds: raw $[0, 1]$ inputs, full training sets, 20 epochs, batch size 64, SGD $\eta = 0.01$ vs. Adam $\eta = 0.001$, measured once at the end (mean $\pm$ s.d.). The "Adam is flatter" result replicates. On CIFAR-10 it is almost entirely the coordinate factor; on MNIST it is a training-loss mismatch.

	CIFAR-10		MNIST	
	SGD	Adam	SGD	Adam
Original metric $\Delta L/L$ (test)	0.034 ± 0.002	0.012 ± 0.003	0.040 ± 0.008	0.014 ± 0.010
Training loss at measurement	1.42 ± 0.05	1.29 ± 0.02	0.136 ± 0.005	0.008 ± 0.004
Test accuracy	0.475 ± 0.016	0.496 ± 0.005	0.957 ± 0.001	0.978 ± 0.002
Hessian trace ($= \operatorname{tr} G$)	1004 ± 31	395 ± 19	96.2 ± 1.2	9.7 ± 3.1
minimum sharpness MS_1 (function)	353 ± 18	320 ± 11	72.0 ± 1.1	8.0 ± 2.4
orbit excess E (coordinates)	2.85	1.23	1.34	1.21
Filter-normalized average, scale 0.1 ($\times 10^4$)	78 ± 34	207 ± 36	41 ± 9	17 ± 7
Multiplicative average, $\sigma = 0.1$ ($\times 10^4$)	109 ± 21	263 ± 13	61 ± 13	11 ± 3

5.8 Convolutional networks

We repeat the audit on the CNN with SGD, SGD+M and Adam (two learning rates each, batch size 128, three seeds; 18 runs). Rescaling now acts on whole channels, and minimum sharpness is computed from the exact per-sample $\operatorname{diag}(G)$ on 512 examples. Fifteen runs reached every loss target. The three Adam runs at the lower learning rate (10^{-4}) reached training loss 0.018–0.034 by the 150-epoch cap, so comparisons at loss 0.01 use only the other three Adam runs.

The mechanism carries over (Fig. 8a). Adam's channel imbalance is roughly twice SGD's in all three hidden layers (median 0.53/0.55/1.47 vs. 0.22/0.35/0.65 at loss 0.1). Its orbit excess stays low at every loss level (median 1.45, 1.78, 1.68 at losses 1, 0.1, 0.01), while SGD's and SGD+M's grow from 1.7–1.8 to 3.2. The coordinate term again favours Adam and grows during training: -0.31 $[-0.38, -0.23]$, -0.71 $[-0.91, -0.49]$ and -0.93 $[-1.06, -0.79]$ bits against SGD, and -0.24 to -0.91 bits against SGD+M.

On this network, however, Adam's solutions are clearly sharper in orbit-minimized sharpness ($+1.27$ to $+2.30$ bits against SGD, $+0.68$ to $+1.24$ against SGD+M; all intervals exclude zero), and every invariant measure rates Adam sharper than both baselines ($P = 0.00$ at every loss for MS_1, multiplicative and $\sum_i w_i^2 G_{ii}$), and so does the partially invariant filter-normalized measure. The coordinate factor therefore does not reverse the ranking here. It *hides* part of the difference. Against SGD+M it hides most of it: the raw trace ratio is $+0.44$, $+0.57$ and $+0.18$ bits with intervals that include zero, and raw measures are inconclusive ($P = 0.25$–0.50 for the trace and the original metric). A researcher using raw measures would conclude that Adam and SGD+M find equally flat minima on this CNN; an invariant measure shows Adam's are about $2\times$ sharper. Across the three settings, the same coordinate bias in Adam's favour thus reverses the ranking (CIFAR-10 MLP), flips it during training (MNIST MLP), or masks a large functional difference (CIFAR-10 CNN), depending on how large the functional difference is.

5.9 Depth

Does the coordinate factor compound with depth, since every hidden layer adds its own rescaling symmetry? We train CIFAR-10 MLPs with $L \in \{1, 2, 4, 8\}$ hidden layers of width 128 with SGD ($\eta = 0.06$), SGD+M (0.006) and Adam (6×10^{-4}), batch size 128 and three seeds (36 runs). All use He initialization with zero biases so that the signal is preserved at every depth, and all reached

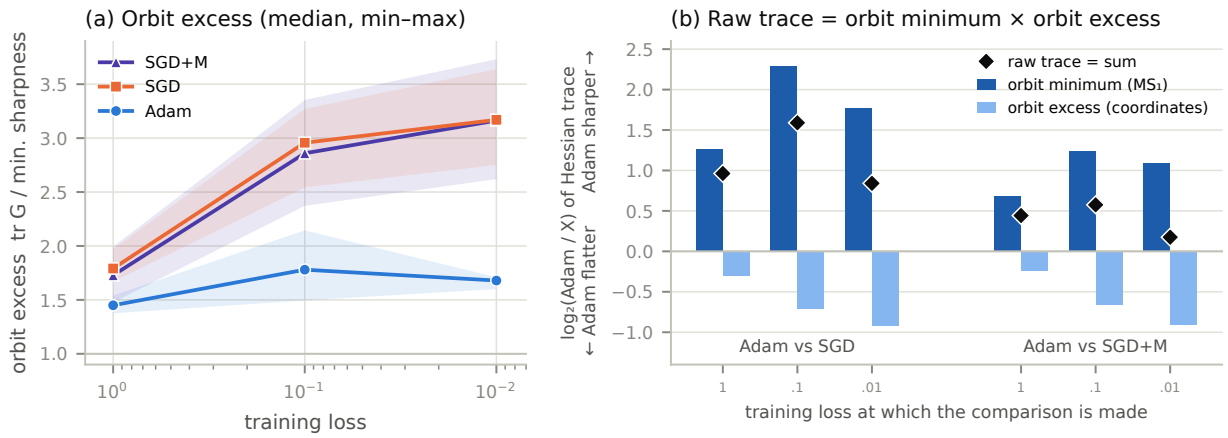


Figure 8: **CNN on CIFAR-10. (a)** Orbit excess at matched training loss; rescaling acts on convolutional channels, through max-pooling and the flatten into the dense layer. **(b)** Exact split of the Adam/X trace ratio. On this network Adam's solutions are sharper in orbit-minimized sharpness, and the coordinate term, which again favours Adam and grows during training, hides most of that difference.

the targets 1, 0.3 and 0.1.[2] The answer is no (Fig. 9). Adam's orbit excess stays at 1.16–1.39 and SGD's and SGD+M's at 1.50–2.15 at every depth. The coordinate term against SGD is −0.48 to −0.74 bits at every depth and target except the earliest target at depth 1 (−0.15), and against SGD+M it is −0.47 to −0.88 bits (again except the earliest target at depth 1, −0.25), a factor of 1.4–1.8 in Adam's favour. What grows with depth is the orbit-minimum term: against SGD it rises from +0.26–+0.37 bits at depth 1 to +0.76–+0.89 at depth 8, and minimum sharpness rates Adam sharper than SGD in every pair at every depth ($P = 0.00$). The consequence mirrors §5.2 and §5.8 within one architecture family. At depths 1 and 2 the raw trace rates Adam flatter than SGD in every pair at losses 0.3 and 0.1 ($P = 1.00$) while minimum sharpness rates it sharper in every pair ($P = 0.00$): the ordering reverses. At depth 8 the raw trace rates Adam sharper as well, but by +0.23 to +0.37 bits instead of the +0.84 to +0.89 bits of the orbit-minimum term, so the coordinate factor hides most of the difference.

5.10 Normalization layers and weight decay

BatchNorm adds a second symmetry. The loss is invariant to scaling each row of a pre-BN weight matrix, so the curvature with respect to that row scales as $1/\|w_{\text{row}}\|^2$, and every optimizer moves along this symmetry: the gradient is orthogonal to a scale-invariant row, so even SGD grows its norm at second order, at a rate set by the learning rate [3, 16, 42]. The ReLU rescaling survives between the BN affine parameters and the next layer. We test whether the coordinate factor still matters with the MLP of §4 with BatchNorm after each hidden linear layer (no bias before BN), trained on CIFAR-10 with SGD ($\eta \in \{0.1, 0.3\}$), SGD+M ($\{0.01, 0.03\}$), Adam ($\{3, 10\} \times 10^{-4}$), SGD+M with weight decay 5×10^{-4} ($\eta = 0.03$) and AdamW (weight decay 0.05, $\eta = 10^{-3}$), batch size 128 and three seeds (24 runs). All runs reached the targets 1, 0.3 and 0.1, the last after a median of 35–44 epochs, and all are within 0.42–0.45 test accuracy. The orbit-minimum machinery does not apply (BN couples the samples), so we compare raw measures with the measures that are

[2] These runs use the reordered CIFAR-10 copy described in §5.5.

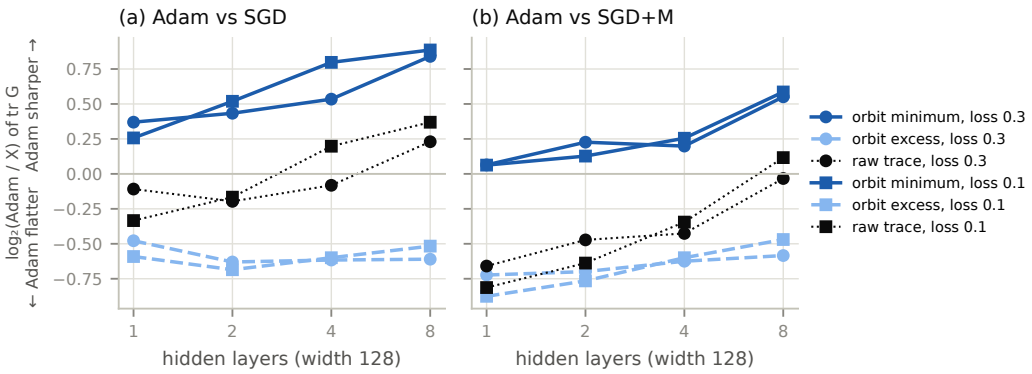


Figure 9: **Depth.** Exact split (2) of the Adam/X trace ratio for CIFAR-10 MLPs with 1–8 hidden layers of width 128 (three seeds each). The coordinate term is about the same at every depth; the orbit-minimum term grows with depth, so the coordinate factor reverses the raw ordering in shallow networks and masks the difference in deep ones.

invariant to every per-weight rescaling: $\sum_i w_i^2 H_{ii}$ (Hutchinson), multiplicative average sharpness and ASAM. All measures use batch statistics.

Raw measures read off the weight norm (Fig. 10). Across the 24 runs, the Hessian trace, $\lambda_{\max}$ and isotropic sharpness are strongly anticorrelated with the median first-layer pre-BN row norm (Spearman $\rho = -0.84, -0.76, -0.62$ for the trace at losses 1, 0.3, 0.1). The invariant measures are not (ρ between -0.30 and $+0.23$ for $\sum_i w_i^2 H_{ii}$ and ASAM). The norms are set mostly by the learning rate. At loss 0.1 the median first-layer row norm is 1.1 for Adam at 3×10^{-4} and 2.9 at 10^{-3}, against 1.3–1.4 for SGD and SGD+M at their smaller rates and 2.2–2.3 at their larger ones. Adam keeps its second-layer rows short at both rates (0.61 and 0.84, against 0.95–1.71 for SGD and SGD+M).

The raw verdict depends on the learning-rate pairing; the invariant verdict does not. Pooled over pairings, raw measures rate Adam *sharper* than SGD ($P = 0.25$ at every target for the trace, $\lambda_{\max}$ and isotropic sharpness; trace ratio $+0.38$, $+0.84$, $+1.00$ bits). Split by pairing, however, Adam at 3×10^{-4} is rated flatter by the trace in 0 of 9 seed-and-target pairs against each of the four SGD and SGD+M settings, while Adam at 10^{-3} is rated flatter in 9/9 against SGD at $\eta = 0.1$ and SGD+M at 0.01, in 4/9 against SGD+M at 0.03, and in 0/9 against SGD at 0.3 (22 of 72 pairs overall). The invariant measures rate Adam *flatter* in every pairing: $\sum_i w_i^2 H_{ii}$ in 70/72 pairs (at least 8/9 per pairing), ASAM in 72/72 and multiplicative sharpness in 64/72. Pooled, P(Adam flatter than SGD) is 1.00, 1.00, 0.92 for $\sum_i w_i^2 H_{ii}$ (-0.27 to -0.47 bits) and 1.00 at every target for ASAM, with the same picture against SGD+M (0.92–1.00). Filter-normalized sharpness, which is invariant to the BN row scaling but not to the BN-affine rescaling, is inconclusive ($P = 0.42$–0.67). In this BatchNorm MLP the coordinate factor therefore again reverses the ordering, but now in the *opposite* direction to the plain MLP: raw measures understate Adam's flatness, because raw curvature is dominated by learning-rate-dependent weight norms rather than by Adam's ReLU-orbit drift. That invariant measures rate Adam flatter here, and sharper on the plain MLP and the CNN, is a further reason not to read our results as a statement about which optimizer finds flatter minima in general.

Weight decay. At the strengths used here, weight decay has little effect before loss 0.1. Compared at the same learning rate, AdamW and Adam have almost the same first-layer norms (2.85 vs. 2.94 at loss 0.1) and raw traces ($+0.04$ to $+0.11$ bits), and SGD+M with weight decay has

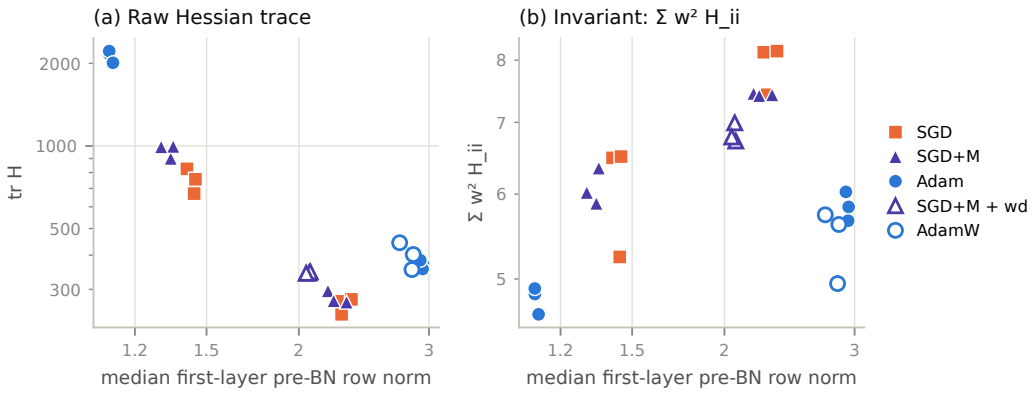


Figure 10: **BatchNorm MLP on CIFAR-10** at training loss 0.1 (one point per run). **(a)** The raw Hessian trace is almost a function of the pre-BN weight norm, which the learning rate sets; Adam at 3×10^{-4} (left) looks sharpest and Adam at 10^{-3} (right) looks about as flat as SGD. **(b)** $\sum_i w_i^2 H_{ii}$, invariant to every per-weight rescaling, does not follow the norm. It ranks every Adam run at 3×10^{-4} below all twelve SGD and SGD+M runs without weight decay, and every Adam run at 10^{-3} below at least nine of them. Hollow markers: with weight decay.

slightly shorter rows than without (2.06 vs. 2.23) and a correspondingly higher raw trace (+0.07 to +0.30 bits) and slightly lower $\sum_i w_i^2 H_{ii}$ (−0.08 to −0.12 bits; three pairs each). AdamW is rated flatter than SGD+M with weight decay by $\sum_i w_i^2 H_{ii}$ and ASAM in all three pairs at every target, while the raw trace again depends on the target ($P = 1.00$, 0.33, 0.00). Stronger weight decay and longer horizons, where the norm equilibrium of van Laarhoven [42] is reached, are tested only in the ResNet-18 standard recipe below.

ResNet-18 on full CIFAR-10. To see whether these observations carry over to a standard architecture, we train ResNet-18, which has BatchNorm after every convolution, on the full 50k-image CIFAR-10 training set (GPU), in two sweeps with three seeds each. The first follows our protocol: no augmentation or weight decay, a constant learning rate, batch size 128, SGD ($\eta = 0.1$), SGD+M (0.01, 0.05) and Adam (10^{-4}, 5×10^{-4}), with checkpoints at clean training loss 1, 0.1 and 0.01 (15 runs). The second is a standard recipe: random crops and flips, a cosine schedule over 60 epochs, SGD+M ($\eta = 0.1$, weight decay 5×10^{-4}), Adam (10^{-3}, L_2 weight decay 5×10^{-4}) and AdamW (10^{-3}, decoupled weight decay 0.05), with the same checkpoints plus the final model (9 runs). Final test accuracy is 82–84% for SGD and SGD+M and 79–85% for Adam in the first sweep, and 94.3–94.8% (SGD+M), 93.4–93.5% (AdamW) and 92.7–93.1% (Adam) in the second. Because every convolution feeds a BatchNorm, the ReLU orbit decomposition does not apply. We use three tiers of measures instead, all computed with batch statistics on 512 clean training images: the raw trace $\operatorname{tr} H$; the trace at unit-norm conv filters (invariant to the BN scale symmetry), which splits $\operatorname{tr} H$ into a term invariant to the BN scale symmetry and a filter-norm term (the split is an identity; the invariance is exact only as BatchNorm's $\epsilon \to 0$, and with $\epsilon = 10^{-5}$ rescaling every filter by up to 3× changes the invariant term by at most 0.3%, App. F); and $\sum_i w_i^2 H_{ii}$ and ASAM, invariant to every per-weight rescaling. The Hutchinson traces use 12 probes (median relative standard error 7–8%). The average-perturbation measures were below their sampling noise at the perturbation sizes we used (median standard error 67–111% of the value), so we do not use them here.

Without augmentation (Fig. 11), the raw trace again follows the filter norms (Spearman $\rho = -0.96$, −0.89, −0.62 at losses 1, 0.1, 0.01), and these are set by the learning rate: at loss 0.01, Adam at 5×10^{-4} has grown its filters to a geometric-mean norm of 1.7, against 0.67–1.04 for the

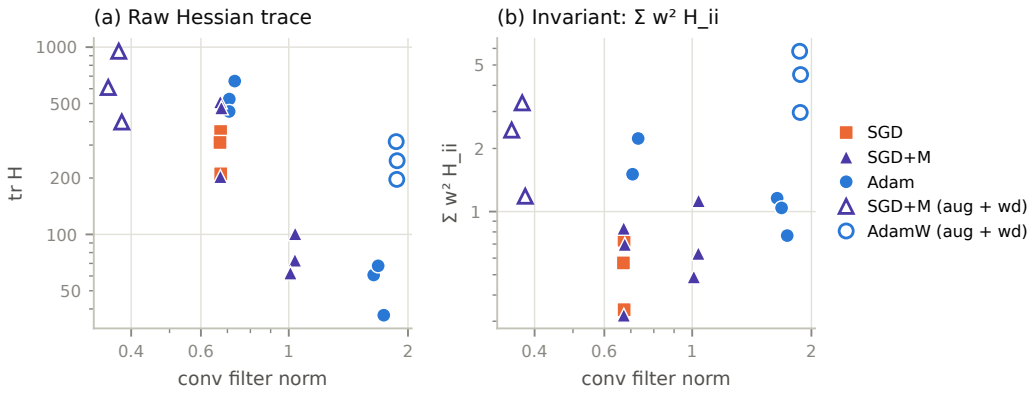


Figure 11: **ResNet-18 on full CIFAR-10** at training loss 0.01 (one point per run; filled: no augmentation or weight decay, hollow: standard recipe with augmentation and weight decay; Adam with L_2 weight decay omitted, see text). **(a)** The raw Hessian trace falls with the conv filter norm, which the learning rate and weight decay set. **(b)** $\sum_i w_i^2 H_{ii}$, invariant to every per-weight rescaling, does not; it rates Adam (filled) sharper than SGD and SGD+M at both learning rates.

other configurations. The raw verdict is accordingly all or nothing per learning-rate pairing. Adam at 5×10^{-4} is rated flatter than SGD and SGD+M at 0.01 in every seed and target, Adam at 10^{-4} almost never (22 of 54 seed-pairing-target triples overall). The per-weight invariant measures give one verdict: $\sum_i w_i^2 H_{ii}$ rates Adam sharper in 53 and ASAM in all 54 triples, by a factor of 2.1–2.9 against SGD for $\sum_i w_i^2 H_{ii}$ (geometric means). At loss 0.01 the exact split of the Adam/SGD trace ratio is -0.75 bits $= +0.14$ (BN-invariant term) -0.89 (filter norms): the norms hide the difference that the invariant measures see. The pattern of the BatchNorm MLP therefore replicates (raw measures read off learning-rate-dependent norms and their verdict depends on the pairing, while invariant measures give one verdict), but the direction of the invariant verdict does not: in ResNet-18, Adam is the sharper one.

With the standard recipe, the raw trace rates AdamW flatter than SGD+M in all three seeds at loss 0.1, at loss 0.01 and at the end of training, by 0.35–2.30 bits (1.3–4.9×). The split attributes most of this to AdamW's larger filters (filter-norm term -1.0 to -1.5 bits; geometric-mean filter norm 1.83–1.88 against 0.34–0.44). The per-weight invariant measures do not support the raw verdict: $\sum_i w_i^2 H_{ii}$ and ASAM each rate AdamW sharper in 7 of these 9 seed-target pairs, some by small margins. With three seeds we report only that the raw advantage is a filter-norm effect, not a reversal. Adam with coupled L_2 weight decay behaves differently again: it drives the filters of the deeper layers to zero, so that in 5 to 8 of the 20 convolutions more than half of the filters have norm below 0.01. This never happens for AdamW or SGD+M and matches the filter-level sparsity reported for Adam with L_2 regularization and BatchNorm [34]. A zero filter feeding a BatchNorm layer makes the curvature depend on BatchNorm's ϵ, so we leave this configuration out of the comparison; the standard-recipe result therefore concerns AdamW against SGD+M only and says nothing about Adam with L_2 regularization. With three seeds per configuration, the ResNet-18 results are preliminary evidence about normalized networks rather than a general claim; App. F gives the full setup.

6 Discussion and limitations

Recommendations. (i) Compare optimizers at matched training loss, not after a fixed number of epochs. (ii) Report at least one rescaling-invariant measure. Minimum sharpness is cheap given $\mathrm{diag}(G)$, and the orbit excess E directly quantifies how much a raw trace comparison owes to the coordinates. (iii) Invariant measures can still disagree with each other (for example the trace at the minimum-norm point and MS_1, which evaluate curvature at different canonical points of the orbit), so state which invariant is used and why. (iv) Use a sanity check such as SAM versus its base optimizer: a measure that cannot tell them apart should not be used to rank optimizers.

What we do and do not claim. We do not claim that Adam's minima are universally sharper. Invariant measures can disagree with each other (§5.2), and sharpness depends strongly on the learning rate. We claim that raw-coordinate flatness comparisons between Adam and SGD-type methods contain an optimizer-dependent coordinate factor, that the factor has a precise mechanism (first-order drift along rescaling orbits, absent for SGD), and that it is large enough to reverse conclusions in the settings we tested.

Scope of each claim. Table 4 lists what kind of evidence supports each claim and the settings it covers. Only the first two rows are mathematical statements; the rest are measurements in the networks we trained, and the explanations attached to rows 6 and 7 are heuristics.

Limitations. Apart from ResNet-18, the experiments are small: MLPs and a small CNN on 10k-example subsets, trained on a CPU. Our theory covers only the ReLU rescaling symmetry. With BatchNorm (§5.10) raw measures are dominated by a second, learning-rate-driven scale symmetry. Our one larger network, ResNet-18 on full CIFAR-10, has BatchNorm after every convolution, so it tests this regime and not the ReLU-orbit decomposition, which remains tested only on networks without normalization; its standard-recipe comparison rests on three seeds. The learning-rate grids are broad but finite. Our explanation of *why* Adam's drift ends near minimum sharpness is a heuristic. Finally, our evidence that the drift helps Adam train (§5.3) comes from one MLP on CIFAR-10. The noise-floor explanation is a heuristic that predicts the direction of the effect, not its size, and whether it matters in larger networks is open.

7 Conclusion

A well-known theoretical caveat, that sharpness depends on the parameterization, has a concrete and systematic practical consequence. Adam breaks the rescaling conservation law that SGD obeys at first order, drifts at a rate $\propto \eta\sqrt{\text{fan-in}}$, and, in the networks we studied, holds its solutions near the flattest representation on their symmetry orbits. The resulting coordinate factor can reverse, flip or mask optimizer flatness rankings, and it weakens sharpness–generalization correlations across optimizers. With BatchNorm a second scale symmetry takes over: raw comparisons then follow weight norms set by the learning rate and weight decay, in a direction that differed between the two normalized networks we studied. Measures invariant to all per-weight rescalings are unaffected by both symmetries (under BatchNorm up to its ϵ). Beyond measurement, the rescaling orbit is a useful probe of optimizer dynamics: it separates a fast, first-order drift of adaptive methods from a slow, noise-driven drift whose destination differs between SGD and Adam as implicit-regularization theory predicts.

Code and data. All code, configurations and per-run results (training curves and all measures at every checkpoint) are available at `https://github.com/Ashvinj2004/LossLandscape-Research`.

Table 4: Claims of this paper, the kind of evidence behind each, and its scope.

Claim	Evidence	Scope
1. First-order orbit drift is zero for SGD, second order for momentum, $\propto r$ for SAM, generically nonzero for diagonal preconditioners	Theorem (Prop. 1); measured	Any network with the rescaling symmetry; loss through the function only; no weight decay; one discrete step
2. $\operatorname{tr} H = \operatorname{tr} G = \mathrm{MS}_1 \times E$	Lemma 2, definition; float64 check	ReLU / max-pool networks without normalization
3. Adam keeps a low orbit excess (1.1–1.8), the SGD family a higher one (up to 4.9)	Measurement	MLPs with 1–8 hidden layers and one small CNN; MNIST and CIFAR-10 subsets; the learning-rate grids tested
4. The coordinate factor reverses, flips or masks Adam–SGD sharpness rankings	Measurement, loss-matched, bootstrap CIs	As row 3
5. Removing Adam's drift doubles raw trace-type sharpness and leaves invariant sharpness unchanged	Intervention (Adam-Q) with controls	CIFAR-10 and MNIST MLPs, 3 seeds; says nothing about generalization
6. The drift helps Adam train in noisy regimes	Measurement; noise-floor heuristic	One CIFAR-10 MLP, 6 of 18 configurations
7. With label noise SGD drifts toward minimum trace, Adam toward minimum $\sum_i \sqrt{G_{ii}}$	Measurement (test of existing theory)	MLPs; 32 label-noise runs per optimizer from 16 independent initializations, plus a frozen-curvature test; per-unit mechanism for SGD only partly confirmed
8. Removing the coordinate factor improves cross-optimizer sharpness–generalization rank correlation	Correlation	Trace-type measures in two MLP sweeps; not causal
9. With BatchNorm, raw measures follow learning-rate- and weight-decay-dependent weight norms; invariant measures give one verdict per network, whose direction differs between networks	Measurement	One BatchNorm MLP (CIFAR-10 subset); ResNet-18 on full CIFAR-10, 3 seeds per configuration

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A Measure definitions

All measures use mean cross-entropy on a fixed evaluation subset of 4096 training examples (CNN: 1024; diag(G) for the CNN uses its first 512), shared by all models. Random perturbations use common random numbers across models.

- *Original metric*: $(\mathbb{E}_\varepsilon L_{\text{test}}(w+\sigma\varepsilon) - L_{\text{test}}(w))/L_{\text{test}}(w)$, $\varepsilon \sim \mathcal{N}(0, I)$ on all parameters, $\sigma = 0.01$, 10 draws, full test set.
- *Isotropic average*: $\mathbb{E}L(w+\sigma\varepsilon) - L(w)$, $\sigma \in \{0.003, 0.01, 0.03\}$, 16 draws; we report $\sigma = 0.01$.
- *Filter-normalized average* [28]: each output unit's (or filter's) row of a Gaussian direction is rescaled to the norm of the corresponding weight row; biases are not perturbed; scale $\in \{0.01, 0.03, 0.1\}$, 16 draws; we report 0.03. Invariant to scaling a row, not to neuron rescaling (§4).
- *Multiplicative average*: $\delta = \sigma|w| \odot \varepsilon$, $\sigma \in \{0.01, 0.03, 0.1\}$; we report 0.03.
- *Hessian trace*: Hutchinson with 20 Rademacher probes. $\operatorname{tr} G$: exact, from the exact diagonal Gauss–Newton matrix; for the MLPs and the CNN it equals the Hessian trace (Lemma 2), of which the Hutchinson value is a stochastic estimate.
- $\lambda_{\max}$: Lanczos (`scipy eigsh`, tolerance 10^{-3}) on Hessian–vector products.
- *SAM worst-case*: $\max_{\|\delta\|_2 \le 0.05} L(w+\delta) - L(w)$ by 10 steps of normalized projected gradient ascent with step 0.4ρ. *ASAM worst-case* [26]: the same with $\|\delta/|w|\|_2 \le 0.5$.
- MS_p: coordinate descent until the relative change is below 10^{-10}. The minimum-norm point: alternating per-unit balancing until $\max_j |\log c_j| < 10^{-7}$.

Verification: the fast MLP diag(G) matches a per-sample implementation to 2×10^{-8}, which matches a brute-force $J^\top S J$ to 3×10^{-8}. Rescalings preserve the network output to 3×10^{-6}. The analytic transport of diag(G) matches recomputation to 2×10^{-7} (MLP) and 3×10^{-7} (CNN). MS_p was never beaten by the raw, minimum-norm or random rescalings tested. Under random rescalings (log-scale s.d. 1), the multiplicative average, ASAM and $\sum_i w_i^2 H_{ii}$ change by less than 2×10^{-4} relative (the residual comes from BatchNorm's ϵ, which moves the loss itself by as much), both for ReLU neuron rescaling in the MLP and for the two symmetries of the BatchNorm MLP (§5.10); filter-normalized sharpness changes by factors of 9.5 and 19 under the neuron and BN-affine rescalings.

B Drift experiment: all optimizers

For sign descent and RMSprop at larger learning rates the second-order term becomes comparable to the first-order one. Each coordinate moves by $\approx \eta$, so $\|u_{a_j}\|^2 - \|u_{c_j}\|^2 \approx \eta^2(d_{\text{in}} - d_{\text{out}})$, which is a second, fan-in-driven contribution in the same direction.

Table 5: Drift experiment (CIFAR-10 MLP, 32×32 inputs, batch size 128, 20 epochs, mean of two seeds). $D^{(1)}$ and $D^{(2)}$ are the cumulative first- and second-order drift terms of (1), averaged over first-layer units; E is the final orbit excess. The decomposition closes to $\leq 1.3 \times 10^{-6}$ in every run.

Optimizer	η	final train loss	$D^{(1)}$	$D^{(2)}$	imbalance (L1)	E
SGD	0.01	1.44	0.0000	0.0023	0.31	1.70
SGD	0.03	0.92	0.0000	0.064	0.32	2.31
SGD	0.1	0.57	0.0000	0.577	0.47	1.78
SGD+M	0.001	1.45	0.0023	0.0001	0.31	1.68
SGD+M	0.003	0.90	0.048	0.0027	0.30	2.41
SGD+M	0.01	0.46	0.576	0.032	0.48	1.99
SAM (SGD+M, $r = 0.05$)	0.003	1.10	0.069	0.0015	0.36	1.80
Adam	3×10^{-5}	1.50	0.067	0.0003	0.41	1.14
Adam	10^{-4}	1.13	0.255	0.0032	0.55	1.14
Adam	3×10^{-4}	0.62	1.076	0.027	0.86	1.12
Adam	10^{-3}	0.27	6.151	0.261	1.40	1.19
RMSprop	3×10^{-5}	1.37	0.114	0.025	0.48	1.11
RMSprop	10^{-4}	0.96	0.397	0.272	0.78	1.10
RMSprop	3×10^{-4}	0.52	1.149	2.186	1.27	1.28
signSGD	3×10^{-5}	1.57	0.052	0.0042	0.40	1.15
signSGD	10^{-4}	1.25	0.149	0.047	0.51	1.14
signSGD	3×10^{-4}	0.81	0.359	0.423	0.78	1.14

Table 6: P(Adam rated flatter than SGD) for each pair of base learning rates, pooled over the 3×3 batch-size pairs, three seeds and training losses $\{0.1, 0.03, 0.01\}$ (81 pairs per cell).

		CIFAR-10				MNIST			
Adam η_0	SGD η_0	$\operatorname{tr} G$	orig.	MS_1	filter	$\operatorname{tr} G$	orig.	MS_1	filter
10^{-4}	0.01	1.00	1.00	0.00	0.02	0.84	0.91	0.42	0.26
10^{-4}	0.03	0.06	0.00	0.00	0.05	0.19	0.58	0.00	0.10
3×10^{-4}	0.01	1.00	1.00	0.27	0.05	0.98	0.93	0.88	0.54
3×10^{-4}	0.03	0.60	0.59	0.07	0.09	0.68	0.83	0.21	0.33

C Robustness to the learning-rate pairing

D Calibration of learning rates

Before the main sweeps we trained every optimizer with four learning rates and three batch sizes for up to 200 epochs (seed 0, no measures). Table 7 reports the epoch at which the training loss first fell below 0.01, or the final loss in parentheses if it did not. No single learning rate trains stably at every batch size: SGD at $\eta = 0.1$, for example, fails at $B = 32$ but works at $B = 512$. We therefore use square-root scaling, $\eta = \eta_0\sqrt{B/32}$, with the base rates $\eta_0 \in \{0.01, 0.03\}$ (SGD), $\{10^{-3}, 3\times10^{-3}\}$ (SGD+M, SAM) and $\{10^{-4}, 3\times10^{-4}\}$ (Adam). Every main-sweep run reached all loss targets within 300 epochs.

Table 7: Calibration: epochs to reach training loss 0.01 (final loss in parentheses if not reached in 200 epochs).

CIFAR-10 MLP					MNIST MLP				
Opt.	η	B=32	128	512	Opt.	η	B=32	128	512
SGD	0.01	67	(0.01)	(0.85)	SGD	0.01	61	(0.02)	(0.16)
SGD	0.03	49	94	(0.10)	SGD	0.03	20	80	(0.03)
SGD	0.1	(0.77)	56	163	SGD	0.1	9	24	98
SGD	0.3	(2.30)	(0.79)	(1.23)	SGD	0.3	19	10	35
SGD+M	0.001	67	(0.01)	(0.86)	SGD+M	0.001	61	(0.02)	(0.16)
SGD+M	0.003	63	80	(0.04)	SGD+M	0.003	21	81	(0.04)
SGD+M	0.01	(0.64)	63	98	SGD+M	0.01	9	26	103
SGD+M	0.03	(2.29)	(0.50)	75	SGD+M	0.03	20	11	36
SAM	0.001	129	(0.05)	(1.02)	SAM	0.001	61	(0.01)	(0.16)
SAM	0.003	105	129	(0.12)	SAM	0.003	23	77	(0.03)
SAM	0.01	(0.32)	93	161	SAM	0.01	12	27	96
SAM	0.03	(1.60)	(0.23)	106	SAM	0.03	13	12	35
Adam	3e-5	(0.02)	(0.41)	(0.91)	Adam	3e-5	161	(0.06)	(0.17)
Adam	1e-4	97	164	(0.13)	Adam	1e-4	50	112	(0.03)
Adam	3e-4	60	81	158	Adam	3e-4	20	44	99
Adam	1e-3	(0.08)	52	70	Adam	1e-3	16	18	38

E Edge-of-stability observation

In pilot runs with the original raw inputs (CIFAR-10 MLP, batch size 128), SGD's $\lambda_{\max}$ tracked the stability threshold $2/\eta$: 17–22 at $\eta = 0.1$ (threshold 20), 47–74 at $\eta = 0.03$ (67), and 187–252 at $\eta = 0.01$ (200). Sharpness comparisons at a single learning rate per optimizer therefore largely compare learning rates.

F ResNet-18 experiment: details

Architecture. The standard CIFAR-10 ResNet-18: a 3×3 convolution with 64 channels and stride 1 (no max-pooling), four stages of two basic blocks with 64, 128, 256 and 512 channels and strides 1, 2, 2, 2, 1×1 convolution projections with BatchNorm on the shortcuts where the shape changes, global average pooling and a linear layer to 10 classes (11.2M parameters). Convolutions have no bias; every convolution is followed by BatchNorm with PyTorch defaults ($\epsilon = 10^{-5}$, momentum 0.1, affine). Default PyTorch initialization; seeds 0, 1, 2.

Data and training. The full 50,000-image training set and the 10,000-image test set, standardized per channel. Batch size 128 (390 steps per epoch), mixed precision (float16 autocast with loss scaling), two NVIDIA T4 GPUs. SGD has no momentum; SGD+M uses heavy-ball momentum 0.9 without dampening; Adam and AdamW use $\beta = (0.9, 0.999)$ and $\epsilon = 10^{-8}$. First sweep: constant learning rate, no augmentation, no weight decay, at most 60 epochs, stopped after the last loss target. Second sweep: random 32×32 crops of the image padded by 4 pixels and random horizontal flips, a cosine schedule over 60 epochs, SGD+M with coupled L_2 weight decay 5×10^{-4}, Adam with coupled L_2 weight decay 5×10^{-4}, and AdamW with decoupled weight decay 0.05 (the same per-step shrinkage $\eta\lambda = 5 \times 10^{-5}$ as SGD+M).

Checkpoints. After every epoch the clean training loss is computed on all 50,000 training images with BatchNorm in evaluation mode; the first epoch at which it falls below 1, 0.1 and 0.01

triggers a measurement. The second sweep also measures the final model.

Measurements. In float32, with batch statistics (BatchNorm in training mode, running averages frozen), on a fixed set of 512 clean training images shared by all runs, in chunks of 128. Hutchinson estimates with 12 Rademacher probes of $\operatorname{tr} H$, of $\sum_i w_i^2 H_{ii}$ and of the trace at unit-norm filters $\sum_i d_i H_{ii}$, where $d_i = \|w_f\|^2$ for the weights of conv filter f and $d_i = 1$ for the BatchNorm and linear parameters. The split $\operatorname{tr} H = \left(\sum_i d_i H_{ii}\right) \times \left(\operatorname{tr} H / \sum_i d_i H_{ii}\right)$ is an identity; the first factor is invariant to rescaling the conv filters when BatchNorm's ϵ is negligible against the pre-BN variances. $\lambda_{\max}$ from 30 Lanczos steps with full reorthogonalization; worst-case SAM ($\rho = 0.05$) and ASAM ($\rho = 0.5$) sharpness by 10 steps of normalized projected gradient ascent. Average-perturbation measures were also recorded but were below their sampling noise and are not used.

Checks. A self-test verifies in float64, with BatchNorm's ϵ set to 10^{-12}, that the loss and the invariant measures are unchanged by filter and ReLU rescalings, that Hessian–vector products match finite differences, and that Lanczos matches a dense Hessian on a narrow network. With the default $\epsilon = 10^{-5}$, rescaling every conv filter of ResNet-18 by a random factor between 0.3 and 3 (log-uniform, the range of filter norms seen in the trained runs relative to initialization) changes the loss by 1×10^{-5} to 1×10^{-4} (relative), the trace at unit-norm filters by 0.2–0.3% and $\sum_i w_i^2 H_{ii}$ by 0.1–0.5%, at initialization and after 30 Adam steps (float64, 64 images, same probes; `scripts/test_resnet_bn_eps.py`). Raw $\operatorname{tr} H$ changes by 120–160% under the same rescaling. The invariances are therefore approximate, with deviations far below the 7–8% sampling error of the trace estimates and the factor-of-two effects we report. The exception is Adam with coupled L_2 weight decay, whose collapsed filters have pre-BN variances comparable to ϵ; it is excluded from the comparisons.